Java Full Stack Development

1. Kafka Basics

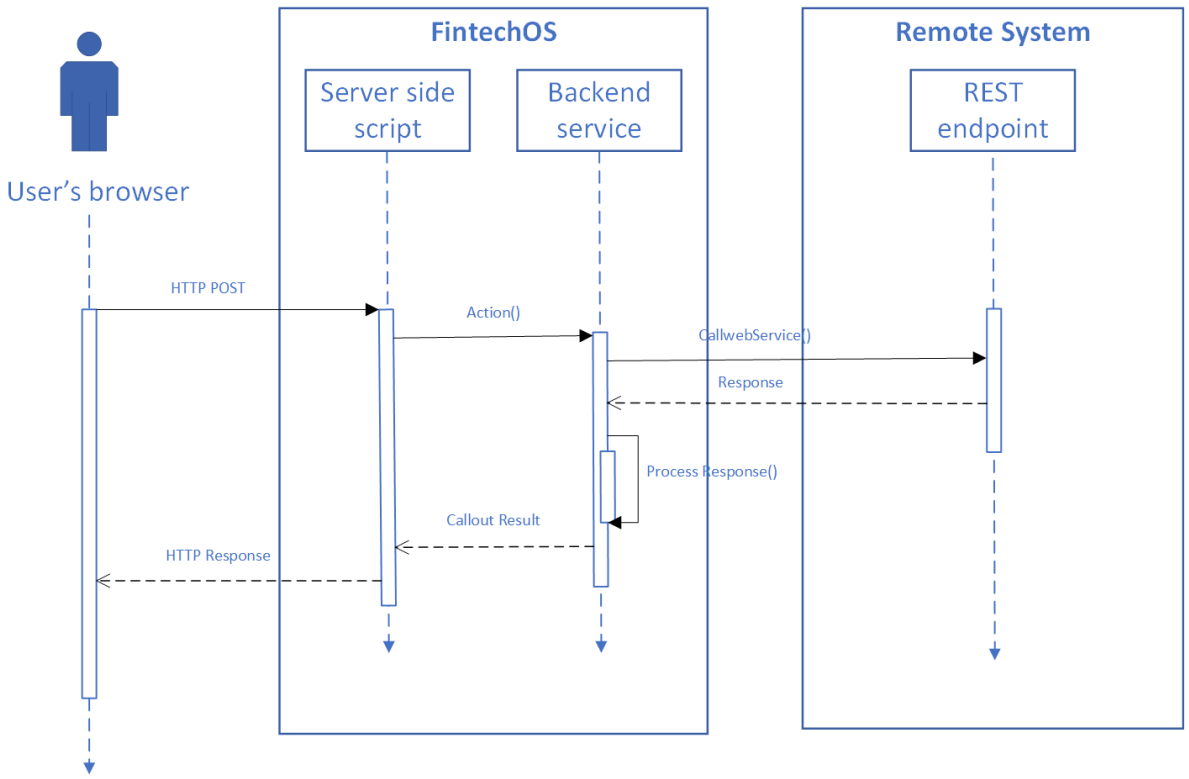
Module Topics

- Event Driven Architecture
- Core Kafka Concepts and Terminology
- Kafka Architecture
- Producers and Consumers
- Design Considerations

Event Driven Architecture

Synchronous Point-to-Point

- Service to service call and response
  - Service A calls Service B over HTTP and waits for a response
  - Simple, familiar, easy to reason about for a single call
- Implicit assumptions baked into this model:
  - B is reachable right now
  - B will respond quickly enough
  - B can handle A's current call rate
  - A has nothing better to do while waiting
- These assumptions hold in small systems
  - They erode rapidly as the system grows



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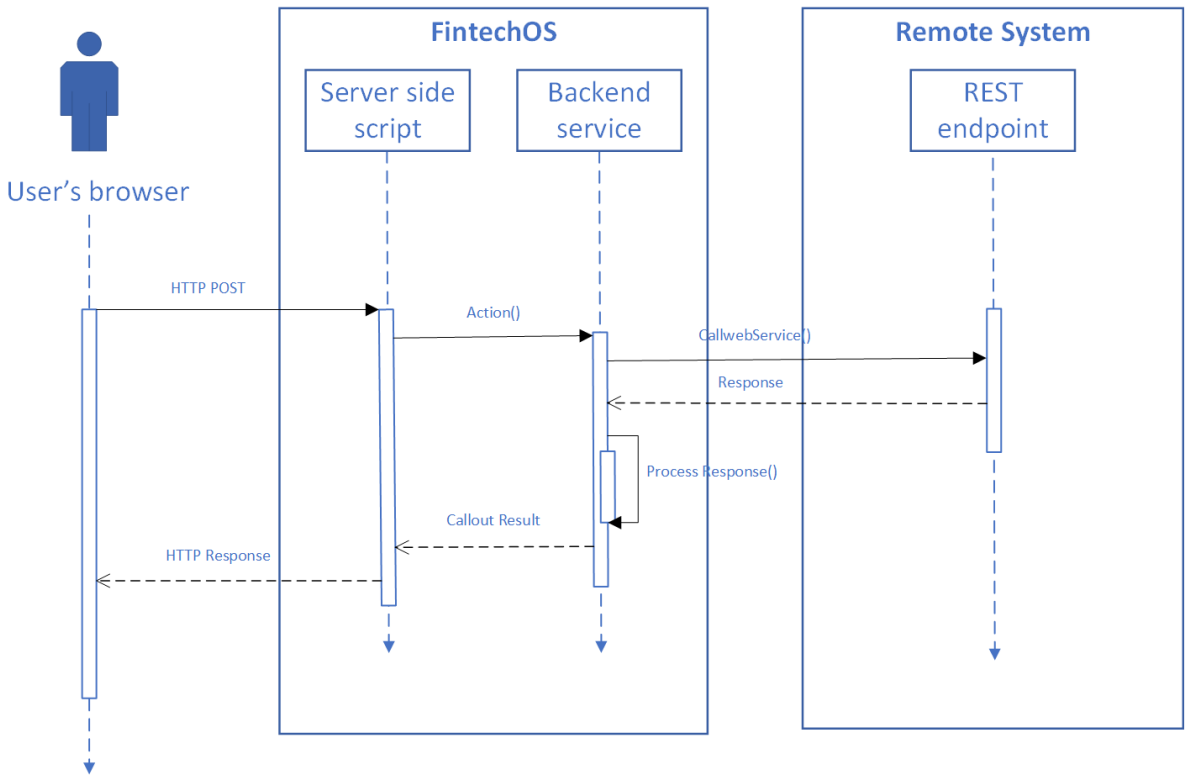
Temporal Coupling Problem

- Synchronous integration requires the caller and callee to be available simultaneously
  - If B is down, A cannot complete its work
  - If B is slow, A is slow
  - This is temporal coupling
    - Services are bound together in time
- Consequences
  - Deployments of B affect availability of A
  - Becomes operationally entangled with it under synchronous integration



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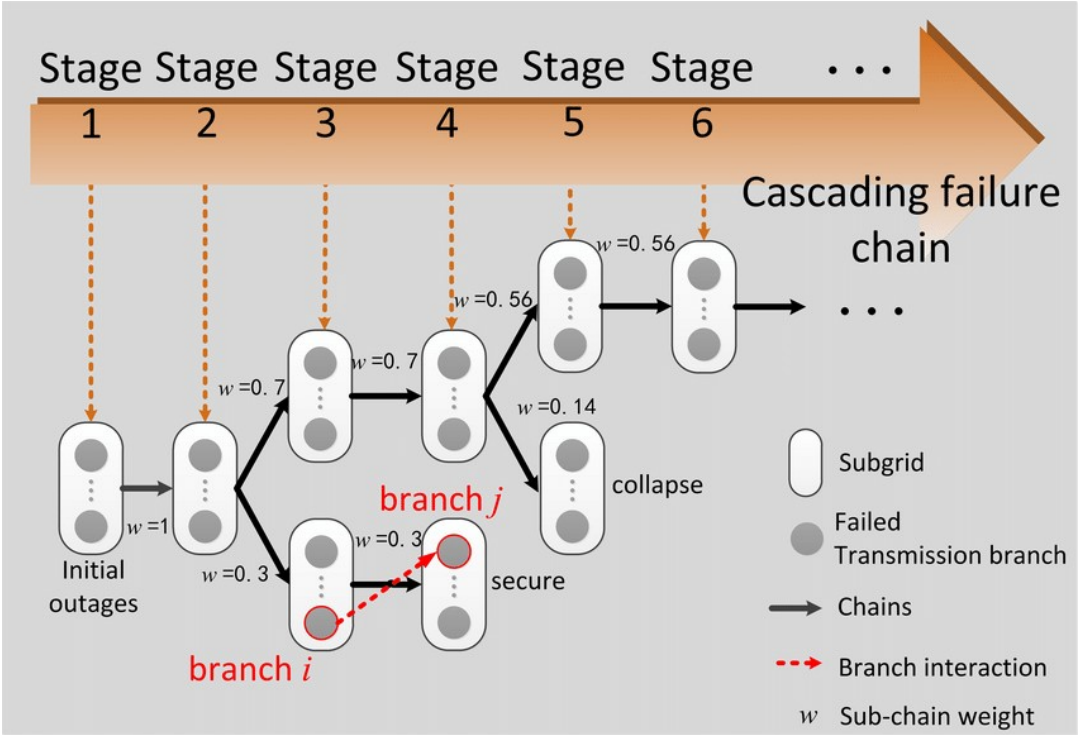
Cascading Failures Problem

- When one service fails, the chain fails
  - A calls B calls C calls D
    - D slows down → C's threads block → B's threads block → A's threads block
    - A single slow downstream dependency can exhaust resources up the entire chain
  - This is the cascading failure pattern
- Common mitigations
  - Timeouts to bound how a wait lasts
  - Circuit breakers to stop calling a known-bad service
  - Bulkheads to isolate resource pools
  - Mitigations only treat the effects of the architecture



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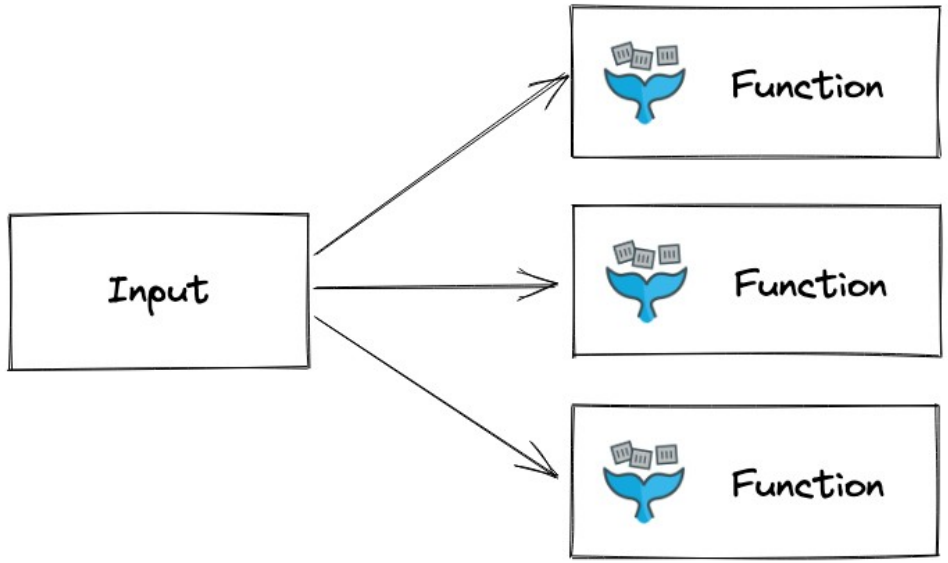
Fan-Out Scaling Failure

- A user updates their profile
  - The Profile service must now notify
    - Search (re-index)
    - Notifications (preference change)
    - Audit (compliance log)
    - Recommendations (refresh model)
    - Analytics (event capture)
  - With synchronous calls, Profile knows about all five consumers
    - Adding a sixth consumer requires changing Profile
  - Profile's latency is the sum of its dependents' latencies
  - Profile's availability is the product of its dependents' availabilities



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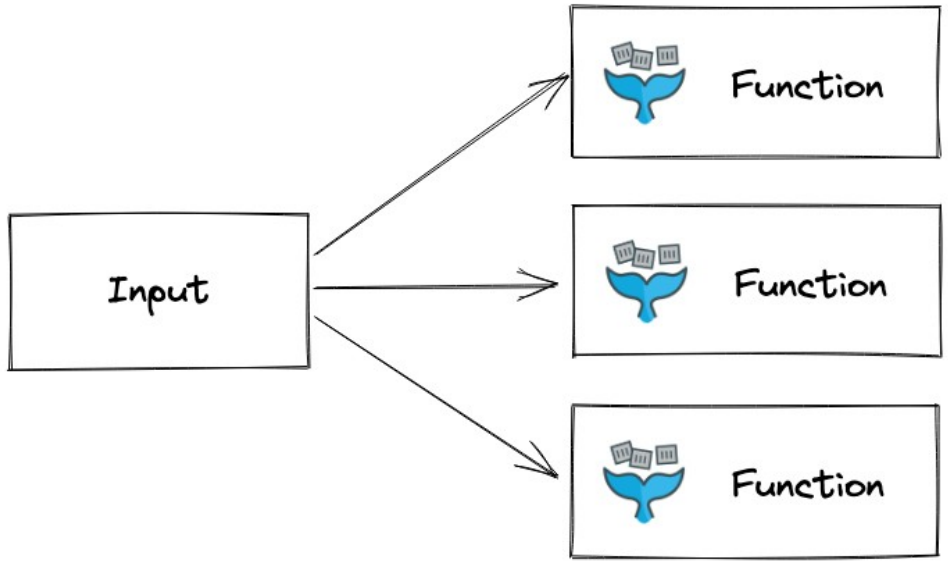
Fan-Out Scaling Failure

- The client becomes a coordinator
  - It has to know
    - Who cares about its events
    - Call each of them
    - Handle each of their failures
    - Pay the latency cost of all of them combined
  - Every new consumer is a code change to the producer
    - Violation of the open/closed principle at the architectural level



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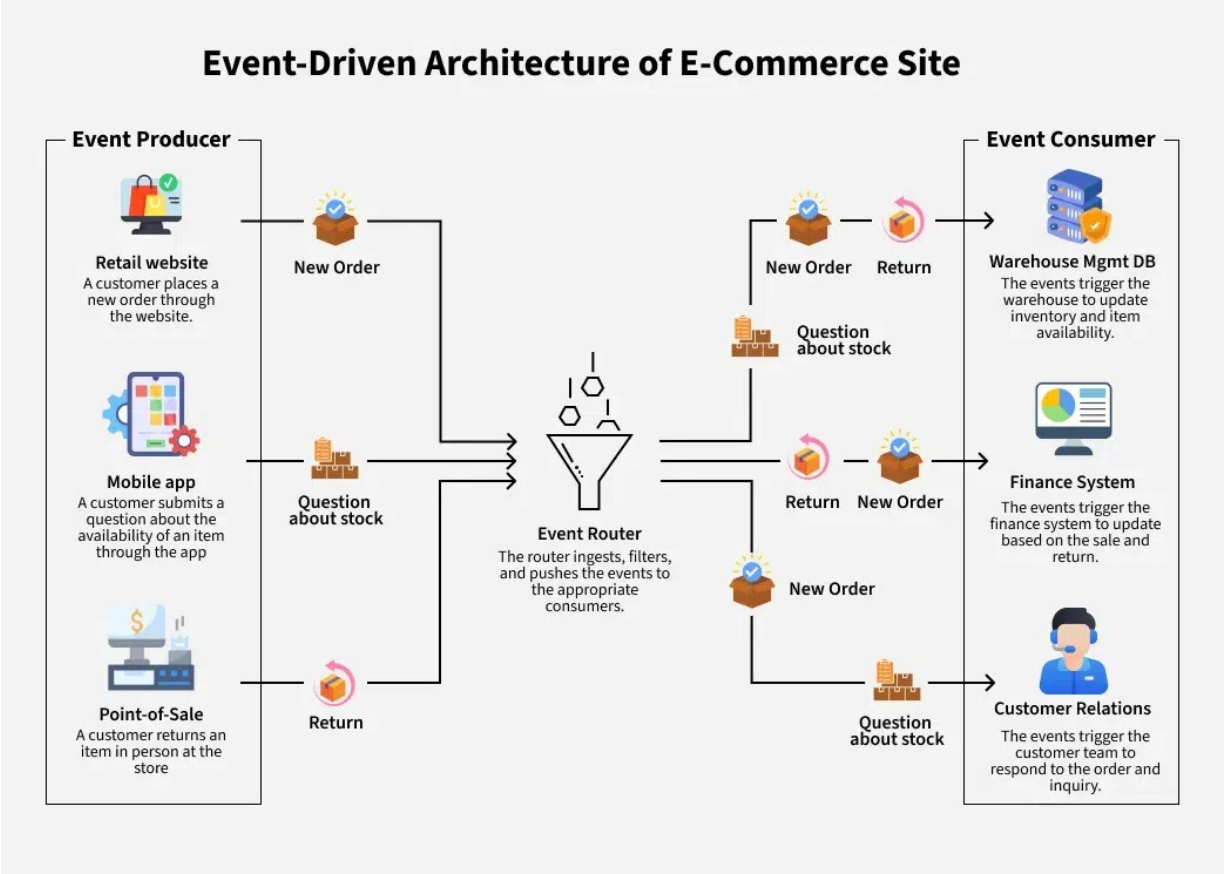
Even Driven Architecture

- In an event-driven system, the primary unit of integration is the event, not the call
  - An event = "something happened, here are the facts"
  - Producers emit events about their domain, without knowing who will consume them
  - Consumers subscribe to event streams, processing on their own schedule
- Communication is usually
  - Asynchronous: no waiting for a response
  - One-to-many: one event, many independent consumers
  - Decoupled in time: producer and consumer need not run simultaneously



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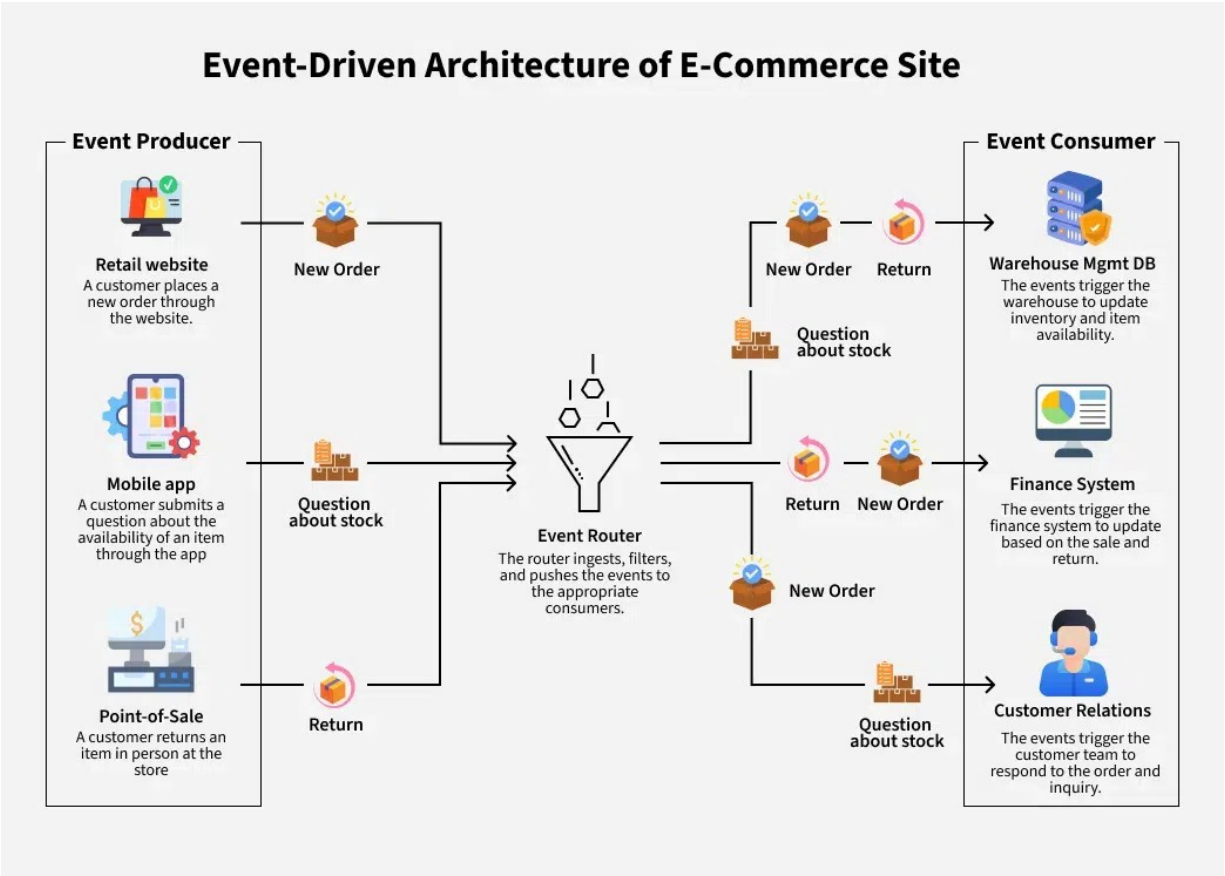
Even Driven Architecture

- In a request/response world
  - Services ask each other to do things
- In an event-driven world
  - Services announce what has happened in their domain, and other services react
  - The producer is no longer responsible for the downstream effects of its events
    - Its job is to faithfully publish what occurred
  - What anyone does with that information is their concern, not the producers



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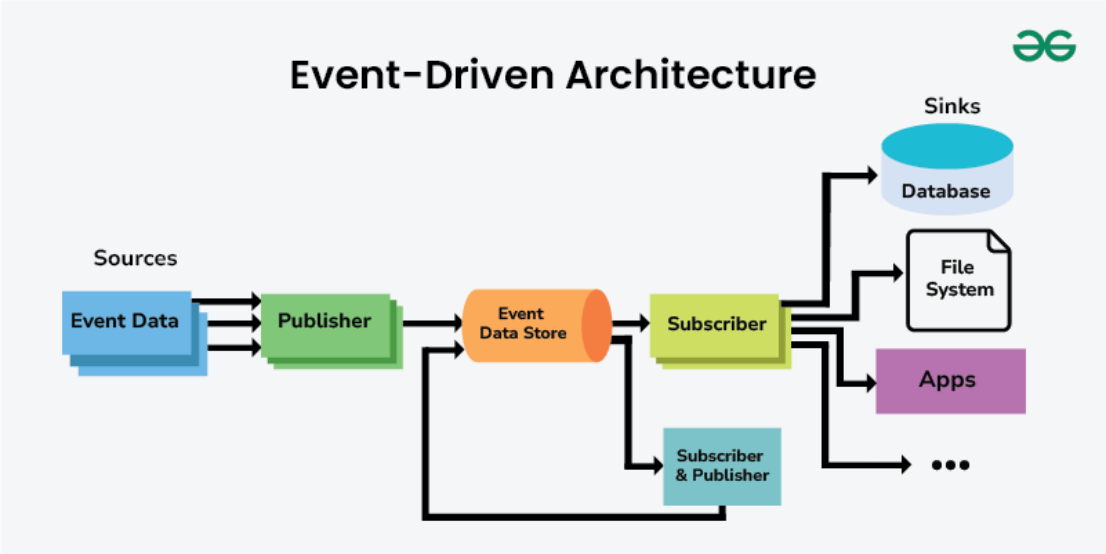
Even Driven Architecture

- Characteristics of event-driven systems
  - Loose coupling
    - Producers don't know consumers; consumers don't know producers
  - Temporal decoupling
    - Services run on independent schedules
  - Replayability
    - Events are durable; consumers can re-process history
  - Auditability
    - The event stream is a record of what happened
  - Extensibility
    - New consumers added without producer changes
  - Natural parallelism
    - Events partitioned across consumers



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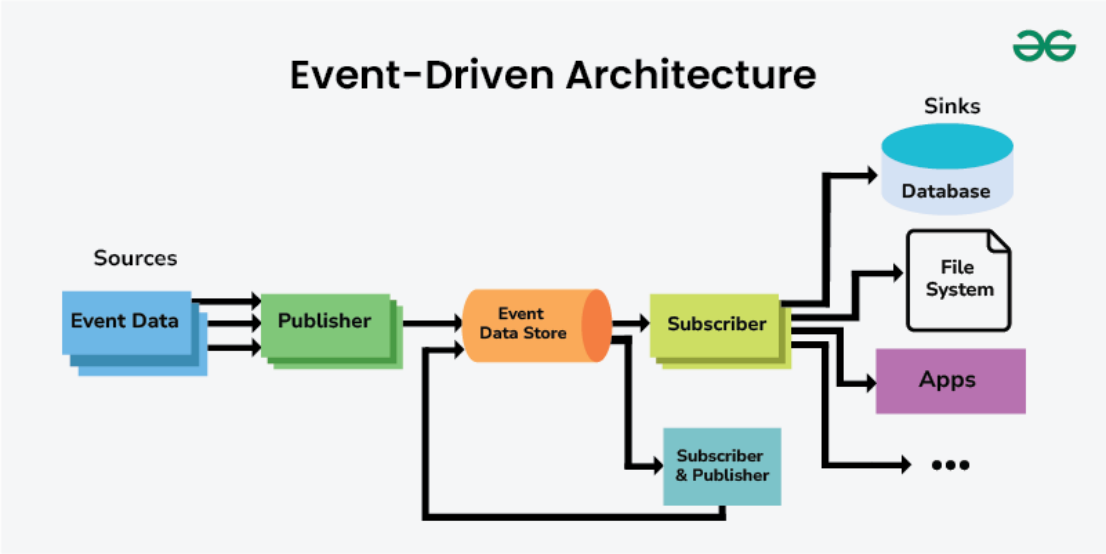
Even Driven Architecture

- Where it works well
  - Workflows where multiple services react to the same domain event
  - Long-running processes that span minutes, hours, or days
  - High-throughput data ingestion and processing pipelines
  - Systems that need an audit trail of state changes
  - Decoupling teams that need to release independently
  - Integration across organizational or team boundaries



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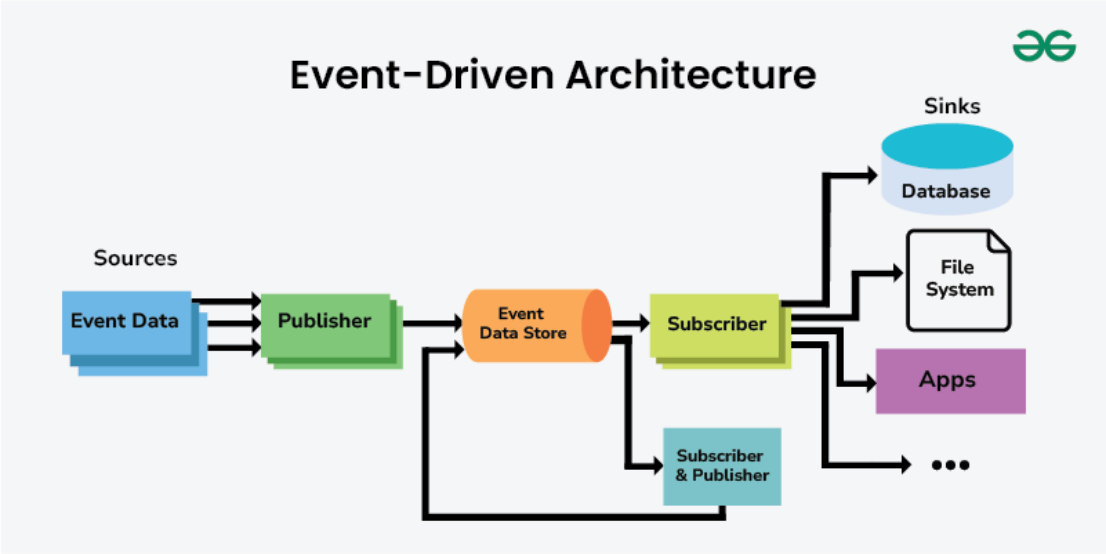
Even Driven Architecture

- Where it should not be used
  - Synchronous user-facing reads where the user is waiting for an answer
  - Strong-consistency operations that must succeed or fail atomically
  - Simple two-service integrations with no fan-out
  - Workflows where the caller genuinely needs the result of the call
  - Operations where eventual consistency is unacceptable to the business



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Core Kafka Concepts and Terminology

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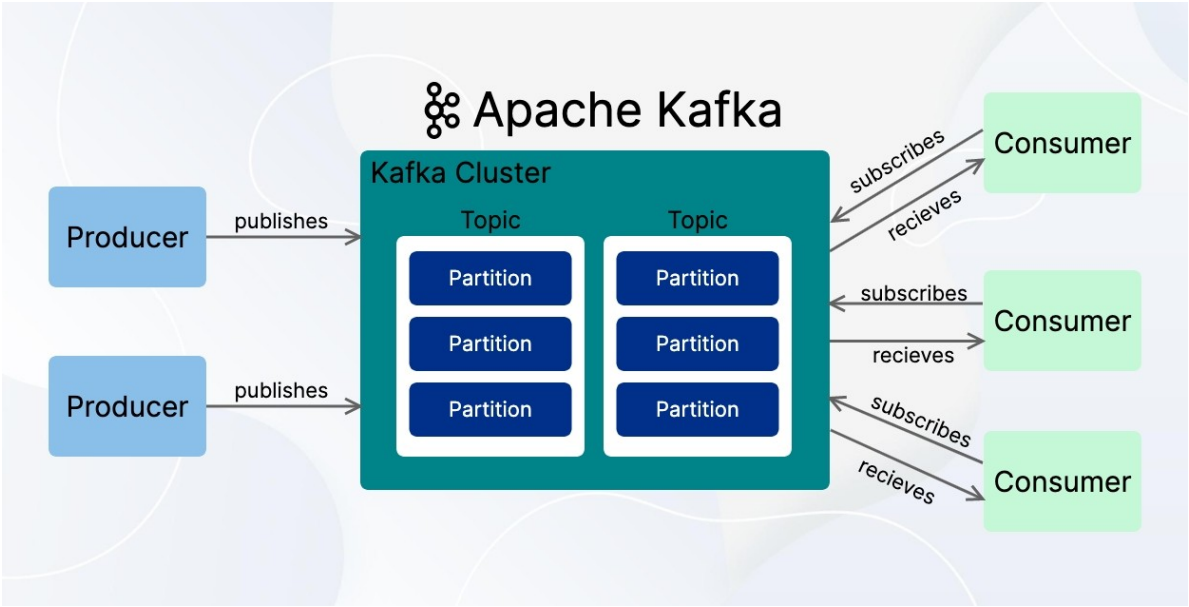
Kafka

- Kafka is a Publisher / Subscriber (Pub-Sub) messaging system
  - Distributed
  - Scales seamlessly
  - High throughput
  - Capable of handling billions of messages per day
  - Replicated
  - Safeguards data in case of machine failures
- Created @ LinkedIn in 2010
  - Now Apache Project (Open Source)



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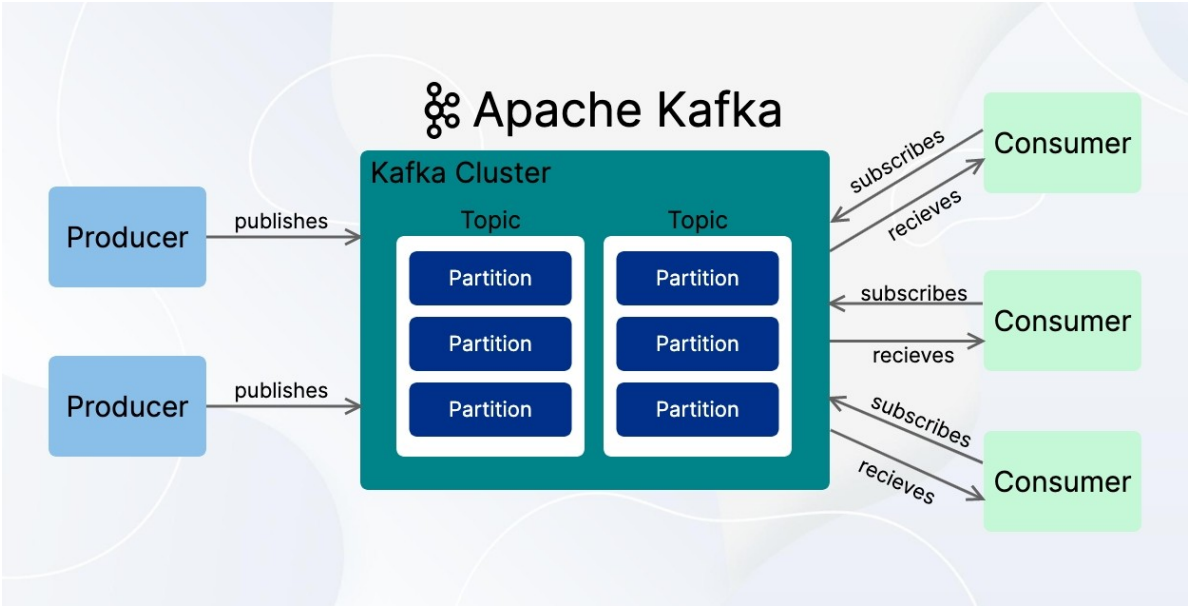
Kafka

- Why LinkedIn Built Kafka?
  - They had lots of databases built to store data
  - Piles of data: relational / key-value / caches / search indexes
- What was missing?
  - Something to handle the continuous flow of data
  - Existing message systems like ActiveMQ, were not able to handle the volume LinkedIn was seeing
  - This motivated LinkedIn to build Kafka from scratch
  - Why name it Kafka?
    - Kafka's co-creator Jay Kreps says "Kafka is optimized for writing" and named after author Franz Kafka



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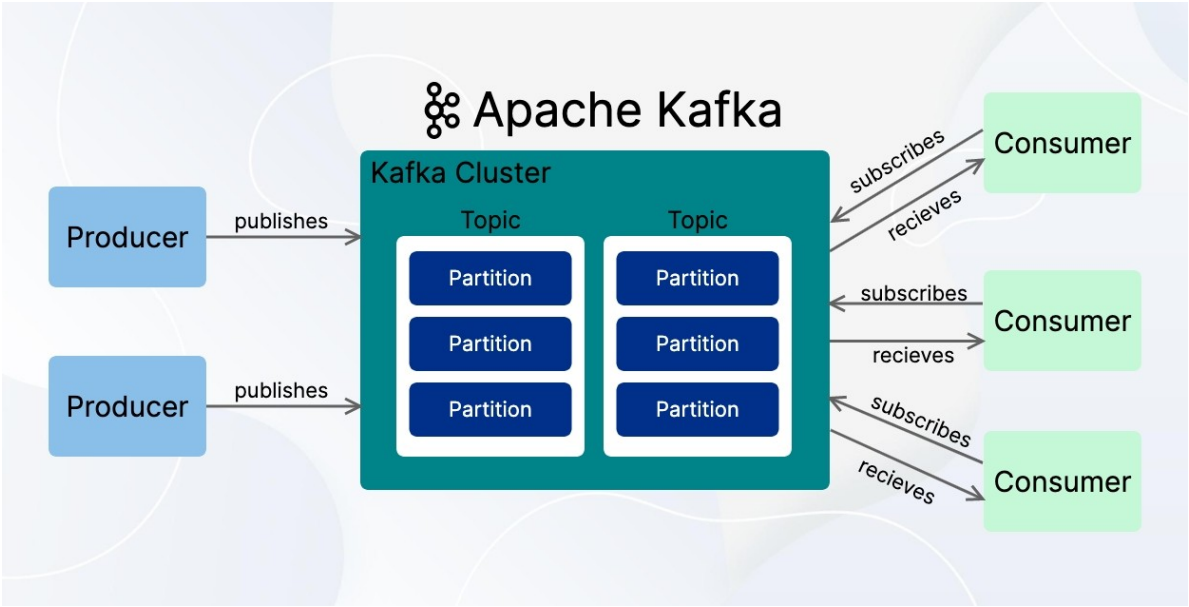
Kafka

- Message deletion
  - In most systems, after a message is 'read' it can be deleted
  - In Kafka
    - Messages are not deleted after they are consumed
    - There is client client API to delete a message
    - Messages are expired by Kafka automatically after a certain time
  - Reason
    - Many applications can subscribe to a topic
    - A client deleting a message can deprive other clients of their input



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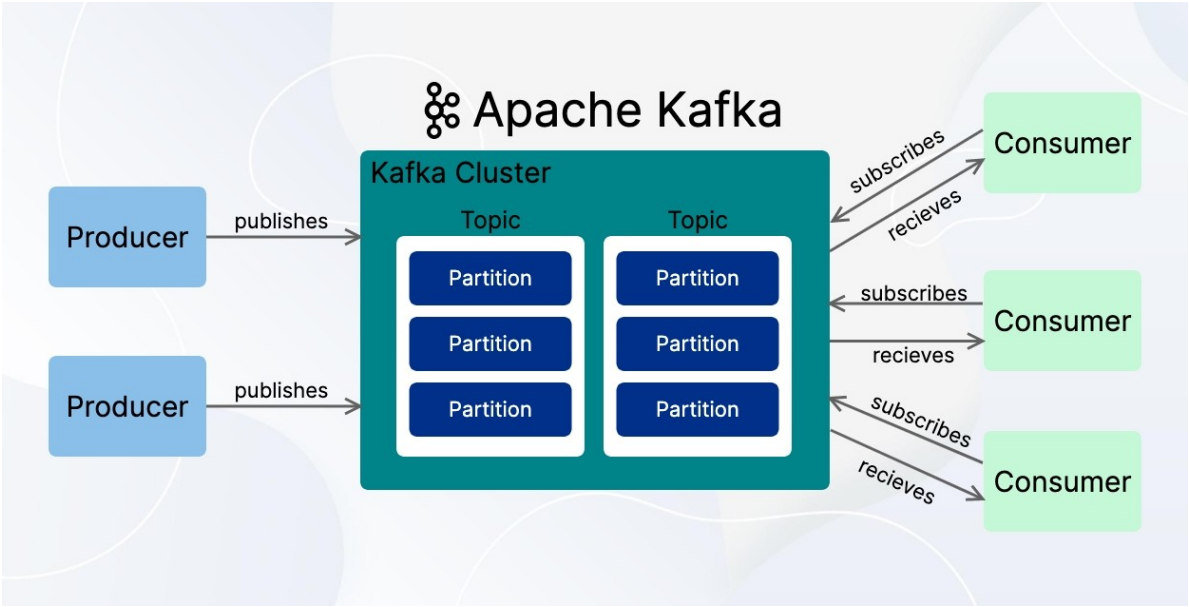
Kafka

- Message read order
  - Other message queues
    - Usually messages are read in order
    - FF and Rewind are not allowed operations
  - Kafka
    - Messages can be read in any order
    - FF and rewind are very fast operations
  - Reason
    - Clients can choose to skip messages if need to be



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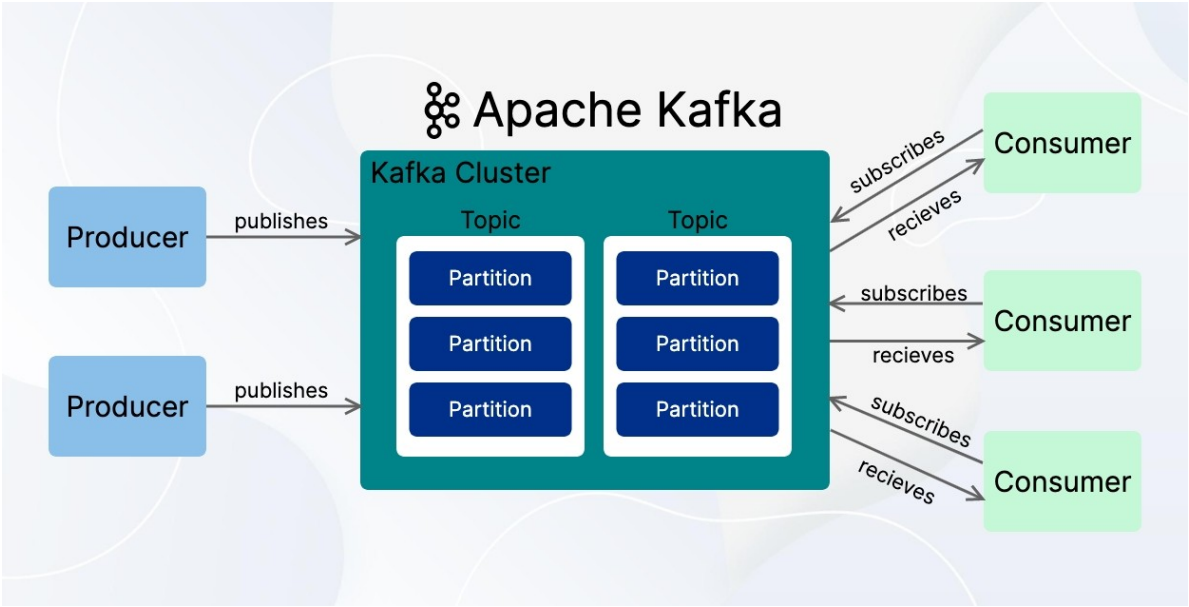
Kafka

- Message processing guarantee
  - Other message queues
    - Hard to guarantee one message will only be processed by one client
  - Kafka
    - Guarantees messages are sent to one consumer
    - No duplicate message processing
    - Uses the mechanism of consumer groups to facilitate this
    - Large number of clients can read / write to Kafka at high speed
    - No locking / blocking



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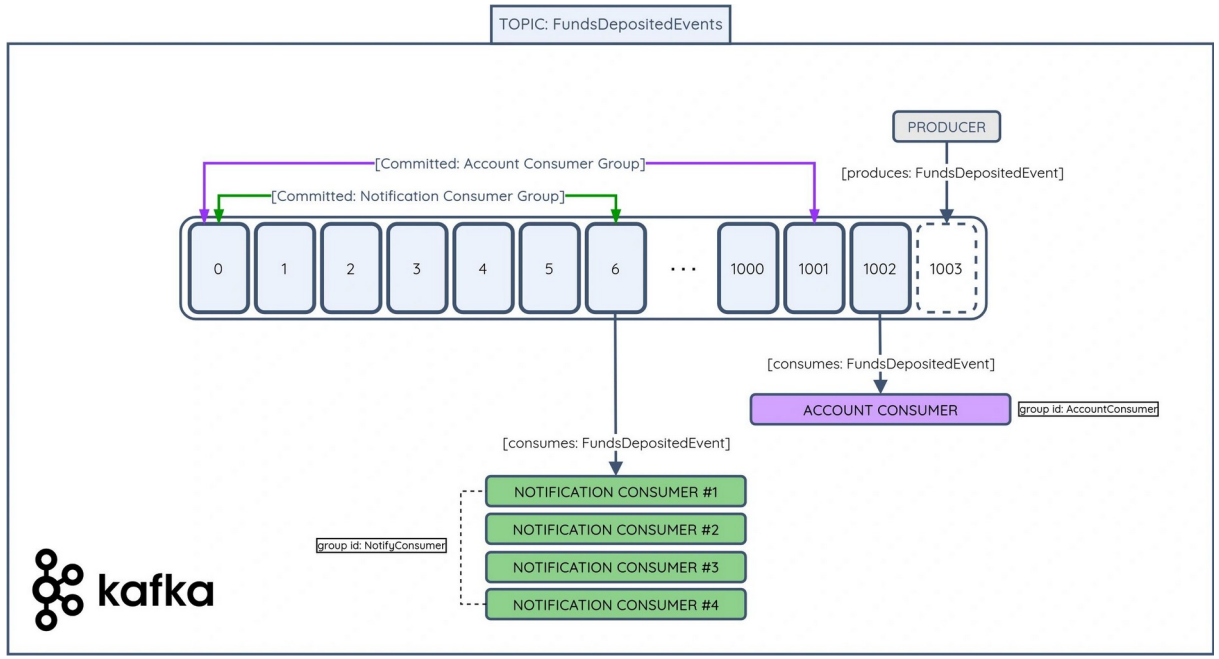
Kafka

- Traditional message brokers
  - Think in terms of queues:
    - Messages arrive
    - Consumers take them
    - Messages disappear
- Kafka
  - Thinks in terms of an append-only log
    - Events are written to the end of a log file
    - Events stay there for a configured retention period
    - Consumers read from a position they choose
    - Many consumers can read the same log independently



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Use Cases

- Messaging and asynchronous service communication
  - The microservices integration use case
- Stream processing and real-time analytics
  - Continuous computation over event streams
- Event sourcing and durable system-of-record event streams
  - The event log is the source of truth
  - State is derived by replaying events

Property	Classical Broker (e.g., RabbitMQ queues)	Kafka
Data model	Queue of messages	Append-only log
Message lifetime	Until consumed	Configurable retention (hours to forever)
Consumer model	Push, broker tracks acks	Pull, consumer tracks offset
Multi-consumer	Requires fan-out exchanges	Native; each consumer group reads independently
Replay	Not supported natively	First-class capability
Throughput per node	Tens of thousands msg/sec	Hundreds of thousands to millions msg sec
Routing logic	Rich (exchanges, bindings)	Minimal (by topic/partition)

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Visualizing the Log

- Events are appended to the right
  - Each event gets a permanent, increasing offset
  - Consumers track their own offset independently
  - No event is "consumed" in the traditional sense, it is read non-destructively

```
Offset:     0    1    2    3    4    5    6    7    8 ...
           +----+----+----+----+----+----+----+----+----+
Events:    | E0 | E1 | E2 | E3 | E4 | E5 | E6 | E7 | E8 | ...
           +----+----+----+----+----+----+----+----+----+
                             ^                   ^
                             |                   |
                   Consumer A position    Consumer B position
                   (reading older data)   (reading recent data)
```

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Visualizing the Log

- A log is a totally ordered, append-only sequence of records
  - This simple structure underlies many distributed systems concepts:
    - Database transaction logs (write-ahead logs)
    - Replication protocols (Raft, Paxos use logs)
    - Filesystem journaling
    - Version control (Git is a log of commits)
  - Logs are easy to reason about because they are
    - Immutable: past entries don't change
    - Ordered: there is a clear "before" and "after"
    - Replayable: you can reconstruct state by reading from the start

```
Offset:     0    1    2    3    4    5    6    7    8 ...
           +----+----+----+----+----+----+----+----+----+
Events:    | E0 | E1 | E2 | E3 | E4 | E5 | E6 | E7 | E8 | ...
           +----+----+----+----+----+----+----+----+----+
                           ^                   ^
                           |                   |
                  Consumer A position    Consumer B position
                  (reading older data)   (reading recent data)
```

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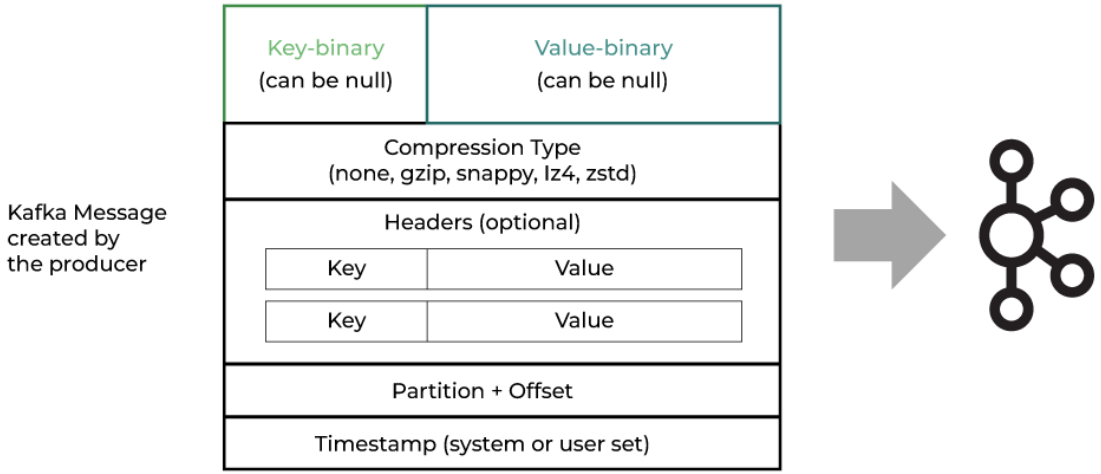
Kafka Events/Messages

- Event
  - A single record written to or read from Kafka
  - Also called a record or message
- Anatomy of an event
  - Key (optional): used for partitioning and ordering
  - Value: the event payload (the "what happened")
  - Headers (optional): metadata key/value pairs
  - Timestamp: when the event occurred or was produced
  - Offset: assigned by the broker; the event's position in its partition



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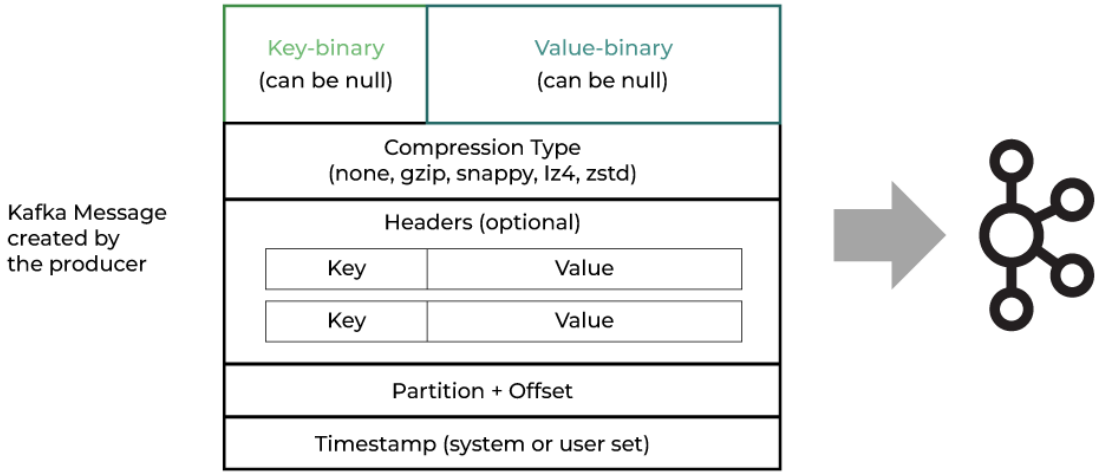
Kafka Events/Messages

- Event
  - Payload is a serialized byte string
  - Has no meaning for Kafka
  - Events are compressed by default
  - Headers provide handling metadata



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Kafka Architecture

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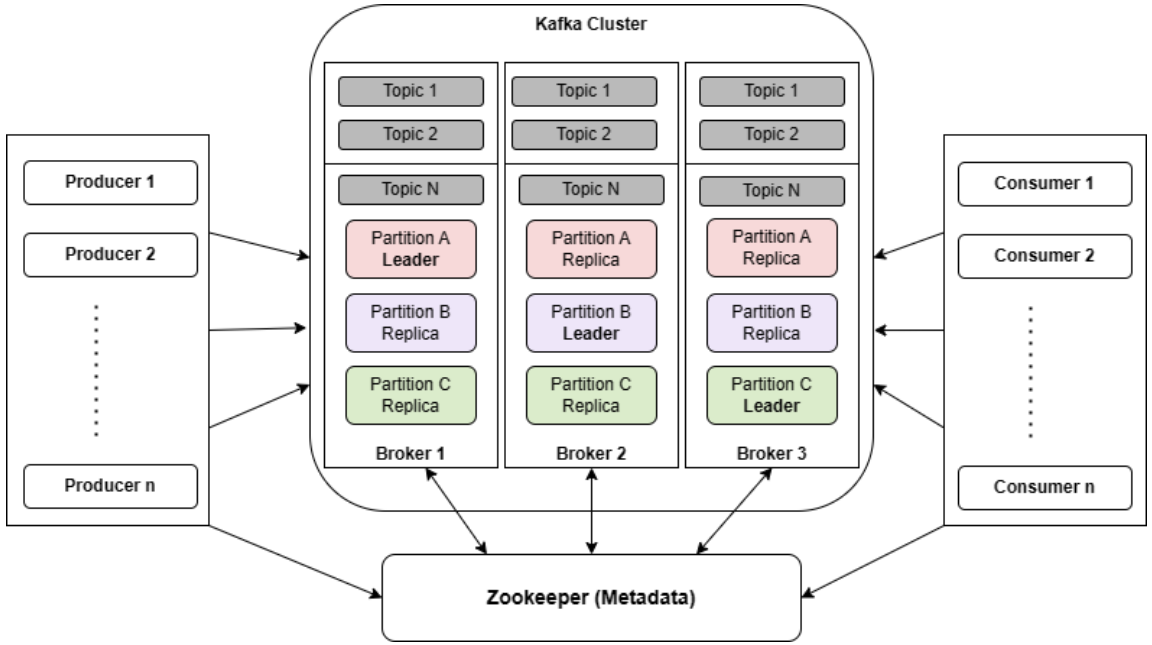
The Topic

- Topic
  - Named stream of related events
  - Examples
    - Orders.placed
    - users.profile-updated,
    - ayments.completed
  - Topics are:
    - Multi-producer: many services can write to the same topic
    - Multi-consumer: many services can read from the same topic
    - Persistent: events are stored on disk for the configured retention
  - A topic is a logical abstraction
    - Physically it is divided into partitions



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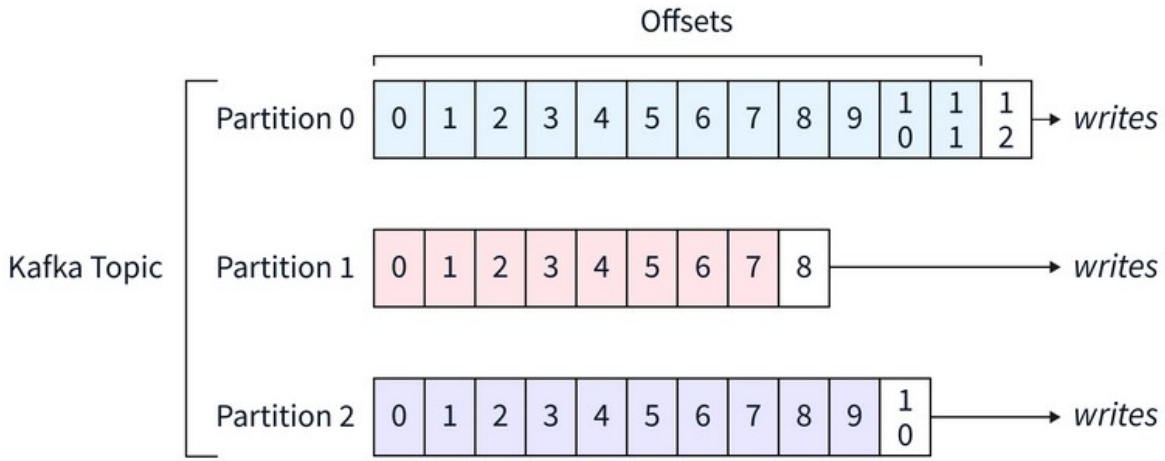
The Partition

- Each partition has its own offset sequence starting at 0
- An event's full address is the triple
  - (topic, partition, offset)
- Three partitions
  - Three independent logs that together form the topic



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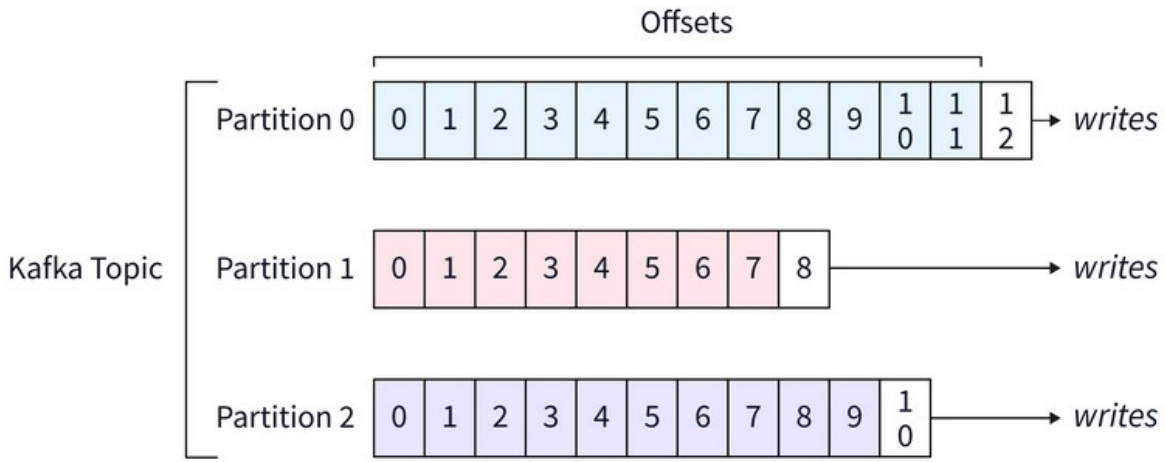
The Offset

- Permanent position marker
  - Monotonically increasing integer assigned by the broker
  - Each partition has its own offset sequence
  - Offsets are
    - Permanent: the offset of an event never changes
    - Per-partition: the same offset value in different partitions refers to different events
    - Consumer-tracked: each consumer remembers where it is in each partition
  - The offset is how a consumer answers: "Where am I in the log?"



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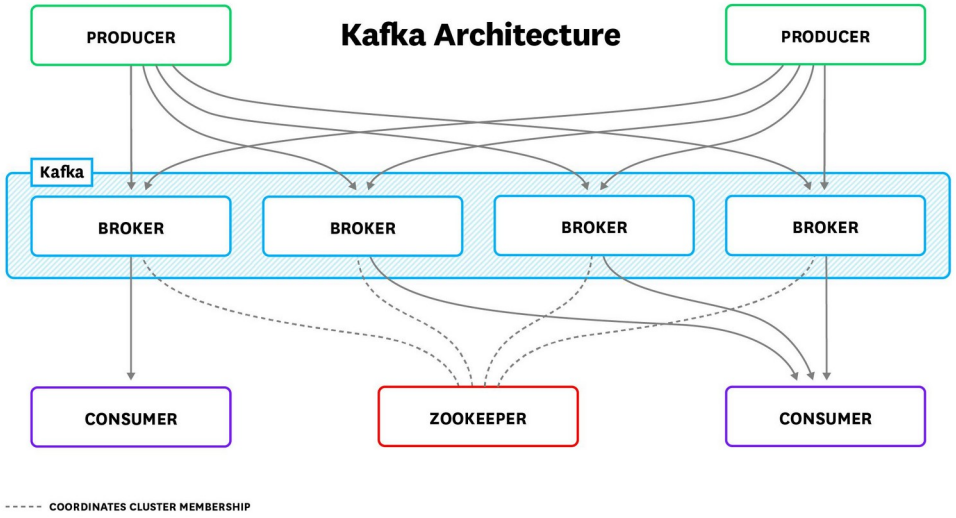
The Producer

- A client that writes events to Kafka topics
  - Producer responsibilities
    - Serialize the event (key and value) to bytes
    - Choose a partition explicitly, or implicitly
    - Send the event to the broker that leads that partition
    - Handle responses, retries, and failures
  - Producers are typically embedded in application services
  - A single application can have one producer that writes to many topics



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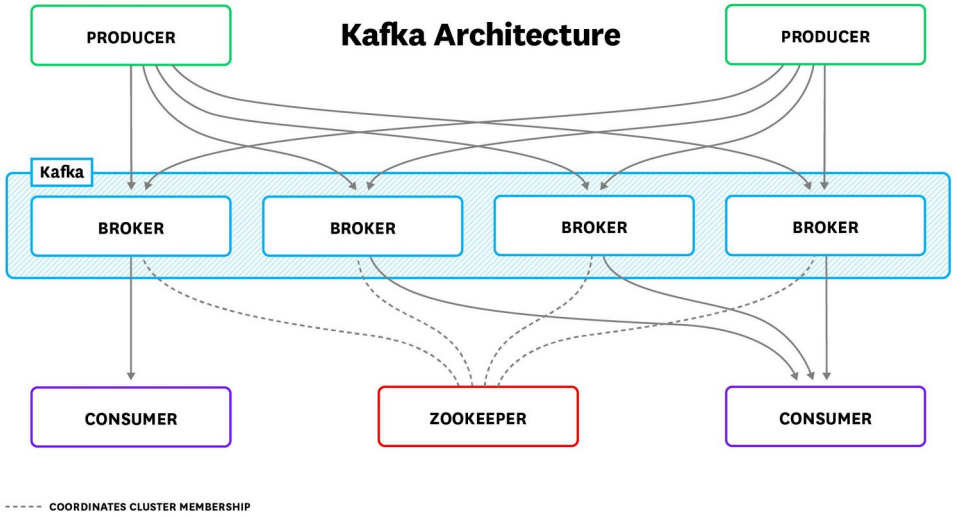
The Consumer

- A client that reads events from Kafka topics
  - Consumer responsibilities
    - Subscribe to one or more topics
    - Receive partition assignments from the broker
    - Poll for events from assigned partitions
    - Process events and commit offsets
  - Consumers pull events at their own rate
  - The pull model means the consumer controls its own throughput



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The Consumer Group

- A set of consumers that cooperate to read a topic
  - Kafka assigns each partition to exactly one consumer in the group
  - Consequences
    - Adding consumers to a group creates more parallelism up to partition count
    - Each event is processed by one consumer in the group
    - Multiple groups can read the same topic independently
    - Prevents consumers in the group from reading the same message twice
  - This is how Kafka delivers both load balancing and fan-out

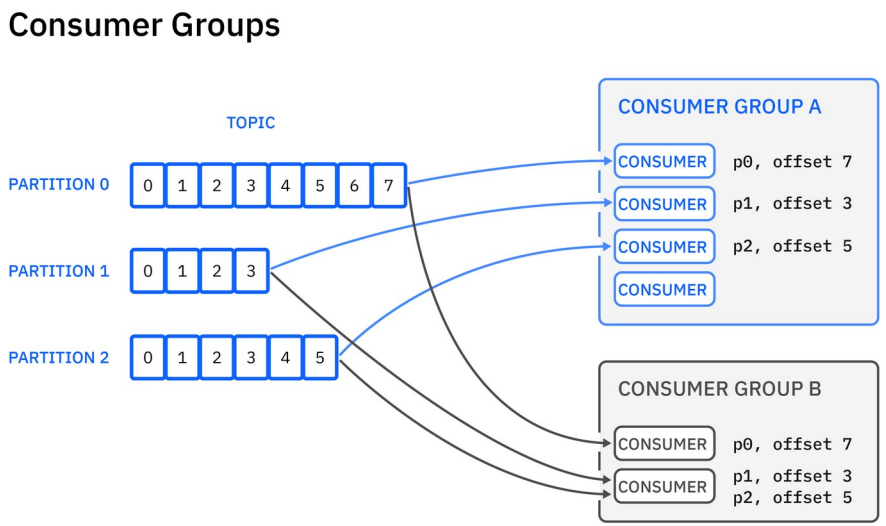

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Image Credit: https://storage.googleapis.com/dctypwgxiwyske/kafka-list-consumer-groups.html

The Consumer Group

- Example:
  - Group “inventory” starts with two consumers registered with Kafka
  - Kafka assigns partitions
    - A1 reads partitions 0 and 2
    - A2 reads partition 1
    - A3 is added, partitions re-balanced
    - Partition 2 is reassigned to consumer A3
    - Consumer B1 has to read from all three partitions
  - Kafka ensures that all partitions in a topic are read by the consumers in a group
    - Avoids the problem of unread messages

```
Topic: orders.placed (3 partitions)

  Partition 0 ──┐          ┌──► Consumer A1   ┐
                ├──────────►│                  │ Group: "inventory"
  Partition 1 ──┤          └──► Consumer A2   ┘
                │
  Partition 2 ──┴──────────► Consumer A3   (Group: "inventory")

  Partition 0 ──┐
                │
  Partition 1 ──┼──────────► Consumer B1   (Group: "analytics")
                │
  Partition 2 ──┘
```

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Retention

- Kafka does not delete events when consumed
  - Events are deleted based on retention policy, configured per topic
  - Two main retention strategies:
    - Time-based retention: keep events for N hours/days (default: 7 days)
    - Size-based retention: keep up to N bytes per partition
  - Whichever limit hits first triggers deletion
  - A third option is log compaction
    - Keep the latest event for each key, indefinitely
    - Useful when the topic represents "current state per entity"

```
Topic: orders.placed (3 partitions)

  Partition 0 ──┐          ┌──► Consumer A1   ┐
                ├──────────►│                  │ Group: "inventory"
  Partition 1 ──┤          └──► Consumer A2   ┘
                │
  Partition 2 ──┴──────────► Consumer A3   (Group: "inventory")

  Partition 0 ──┐
                │
  Partition 1 ──┼──────────► Consumer B1   (Group: "analytics")
                │
  Partition 2 ──┘
```

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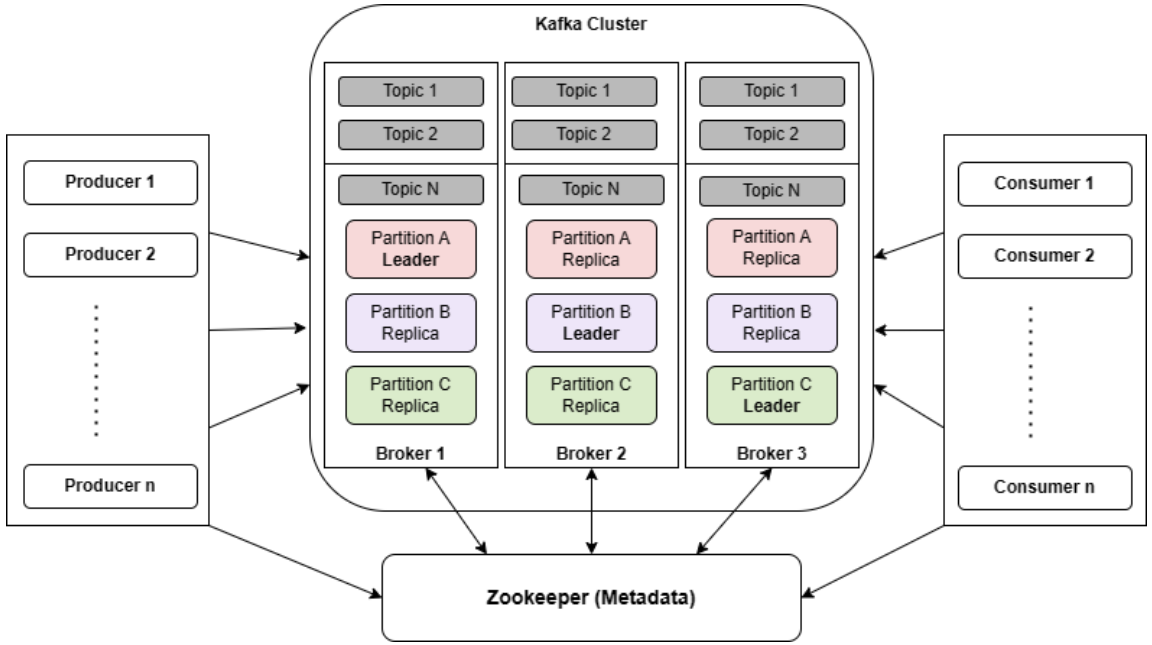
The Broker

- A broker is a single Kafka server process
  - Each broker
    - Hosts a subset of the cluster's partitions
    - Serves read and write requests for those partitions
    - Replicates partitions to and from other brokers
  - A broker is typically one machine
    - Either physical or virtual
- A production cluster has many brokers
  - Typically 3 to dozens or hundreds
  - Cluster metadata is managed by Zookeeper
  - There is no controller “node”
  - One broker is designated arbitrarily as the leade



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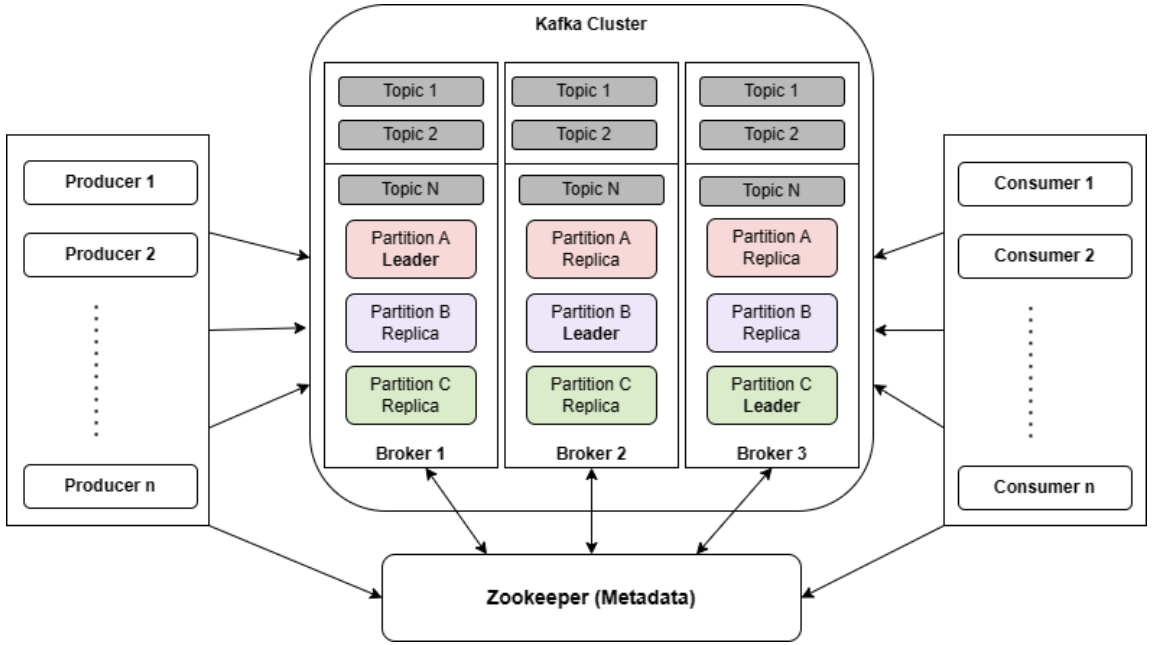
The Cluster

- A cluster is a set of brokers operating together
  - Each partition has a configurable replication factor
- For each partition
  - One broker holds the leader replica that handles all reads and writes
  - Other brokers hold follower replicas
    - These are kept in sync with the leader
  - If a broker where the leader is stored fails
    - One of the followers is promoted to leader
  - This is what gives Kafka its durability guarantee



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The Distributed Log

- A partition
  - Is a directory on the broker's filesystem
  - Inside the directory, the log is split into segment files
  - Only one segment is active, meaning being written to at a time
  - Older segments are immutable
    - They are never modified, only deleted
  - Each segment has companion index files for efficient lookup

```
orders.placed-3/
  ├── 00000000000000000000.log     (old segment, closed)
  ├── 00000000000000000000.index
  ├── 00000000000005000000.log     (old segment, closed)
  ├── 00000000000005000000.index
  ├── 00000000000009847000.log     (active segment)
  └── 00000000000009847000.index
```

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Performance

- Traditional databases require random disk I/O:
  - B-tree updates touch scattered pages
  - Random seeks are slow on any storage medium
- Kafka only ever appends to the active segment:
  - Writes are sequential
    - Sequential disk I/O is orders of magnitude faster than random I/O
    - On modern SSDs: random is faster, but sequential is still faster
    - On spinning disks: the difference was dramatic
  - Reads are also largely sequential
    - Consumers scan forward from an offset

```
orders.placed-3/
  ├── 00000000000000000000.log     (old segment, closed)
  ├── 00000000000000000000.index
  ├── 00000000000005000000.log     (old segment, closed)
  ├── 00000000000005000000.index
  ├── 00000000000009847000.log     (active segment)
  └── 00000000000009847000.index
```

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Page Cache

- Kafka
  - Does not maintain an in-memory cache of events
  - Relies on the operating system's page cache
  - When events are written
    - They land in the page cache first, then flush to disk
  - When consumers read recent events
    - The data is often still in the page cache: no disk I/O
  - Benefits:
    - No duplicate caching
    - Cache survives broker restarts because the page cache is a kernel structure
    - The OS has excellent cache management algorithms

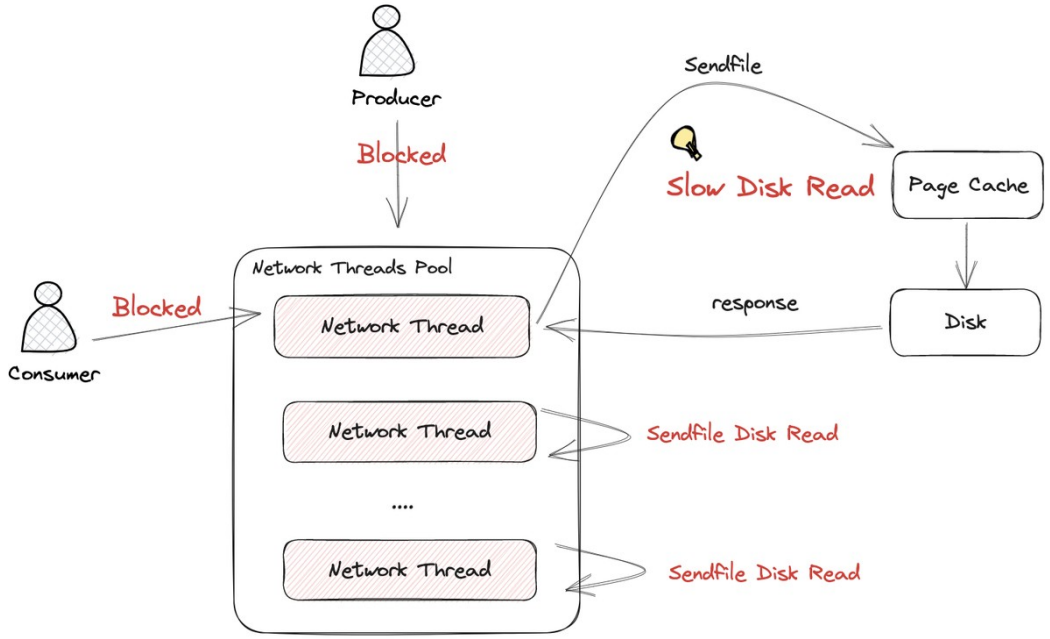

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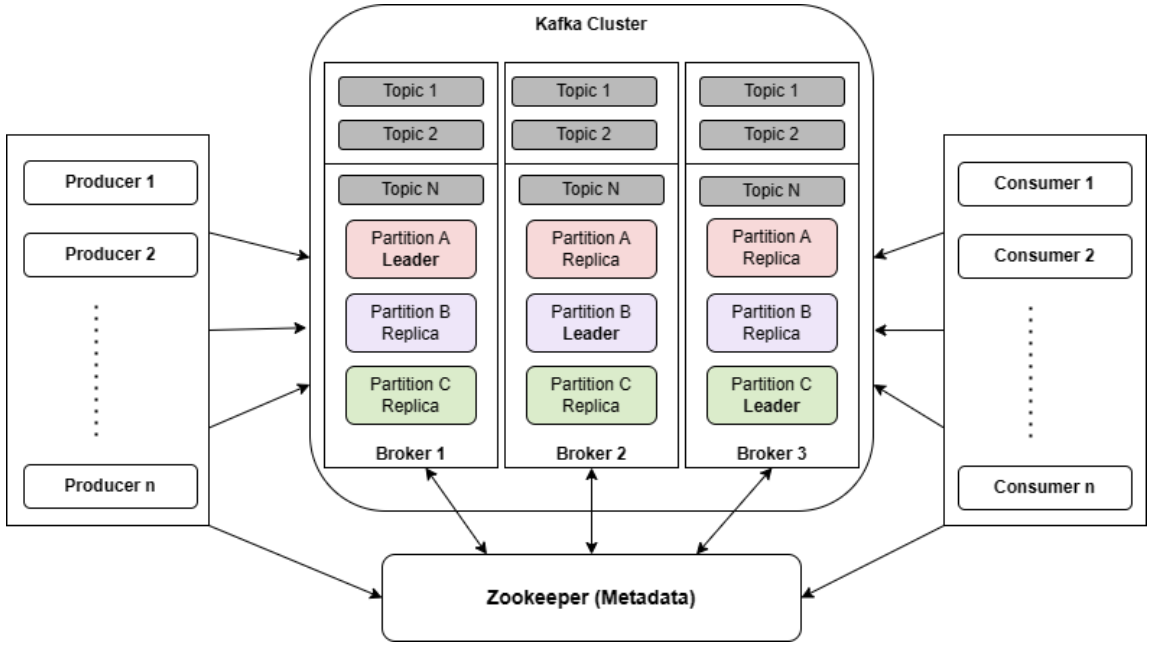
Image Credit: https://github.com/AutoMQ/automq/wiki/Kafka-Design:-Page-Cache-&-Performance

The Broker

- Broker's responsibilities
  - Host a subset of the cluster's partitions
  - Process producer write requests for partitions it leads
  - Process consumer fetch requests for partitions it leads
  - Replicate data to and from other brokers for partitions it follows
  - Report health and metadata to the cluster controller
  - Participate in leader elections when other brokers fail



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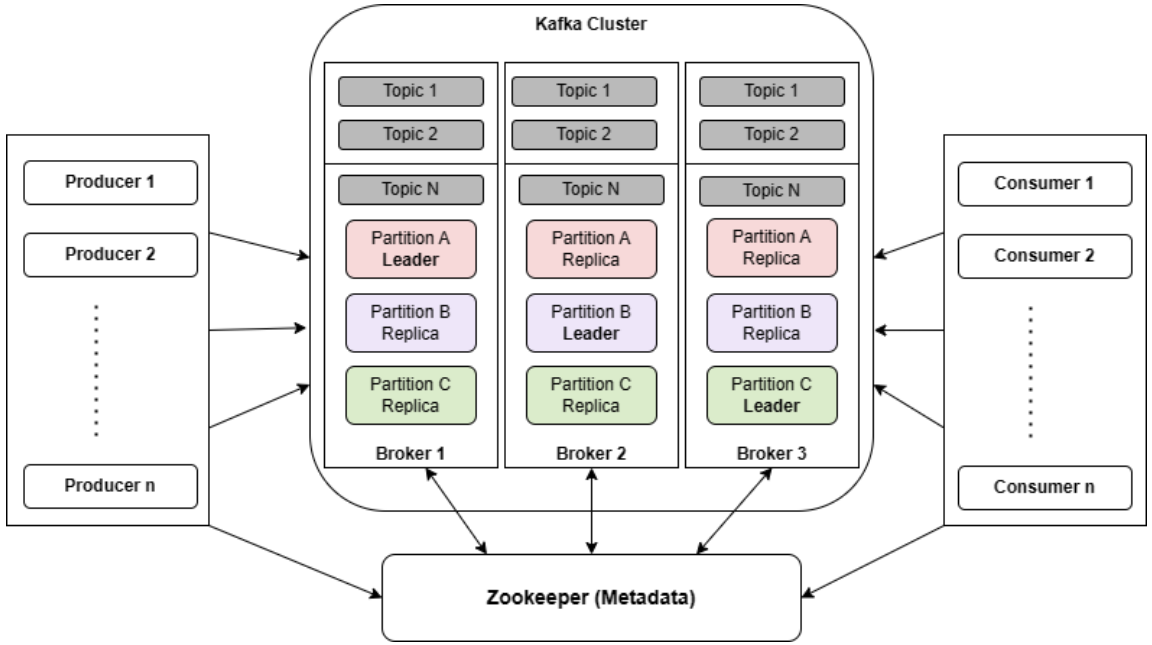
The Cluster

- Brokers presented as one service
  - Clients connect to any broker to discover cluster metadata
  - Metadata includes:
    - List of brokers
    - List of topics and partitions
    - Which broker is the leader for each partition
  - Clients then connect directly to the relevant broker for each partition
    - From the client's perspective, the cluster is a single logical service
  - Internally, the cluster is a coordinated set of peers



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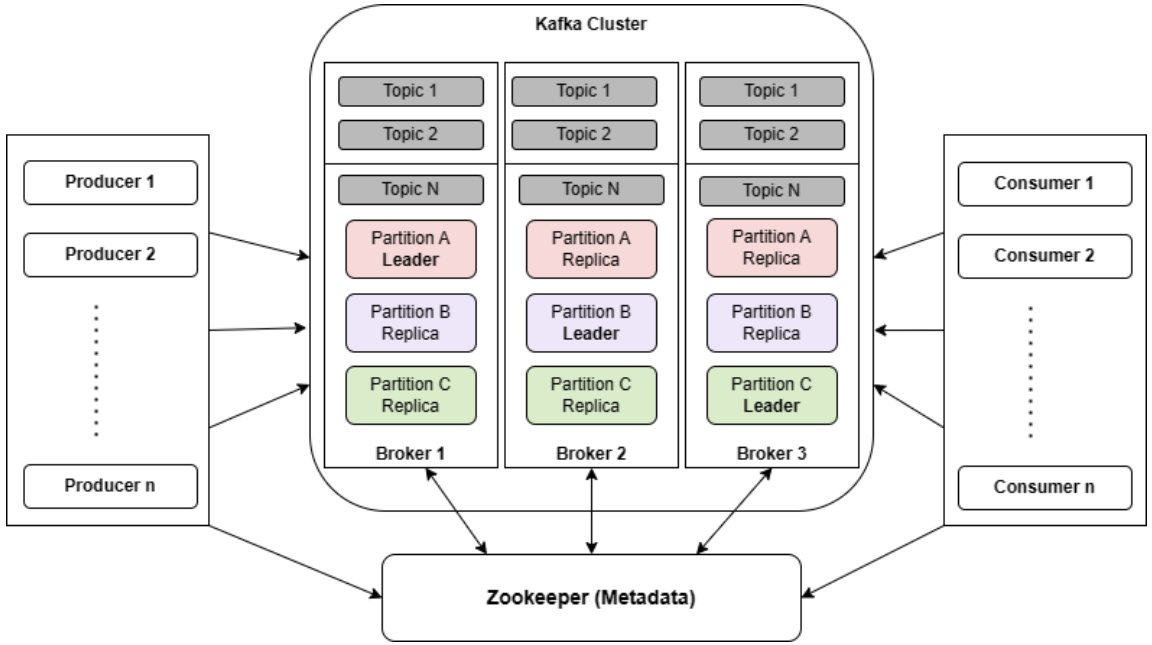
The Controller

- A single broker elected to coordinate cluster-wide decisions
  - Controller responsibilities:
    - Track broker liveness (who is up, who is down)
    - Manage partition leader assignments
    - Trigger leader elections when needed
    - Handle topic creation, deletion, and configuration changes
  - Every Kafka cluster has exactly one controller at a time
  - If the controller dies, another broker is elected to replace it



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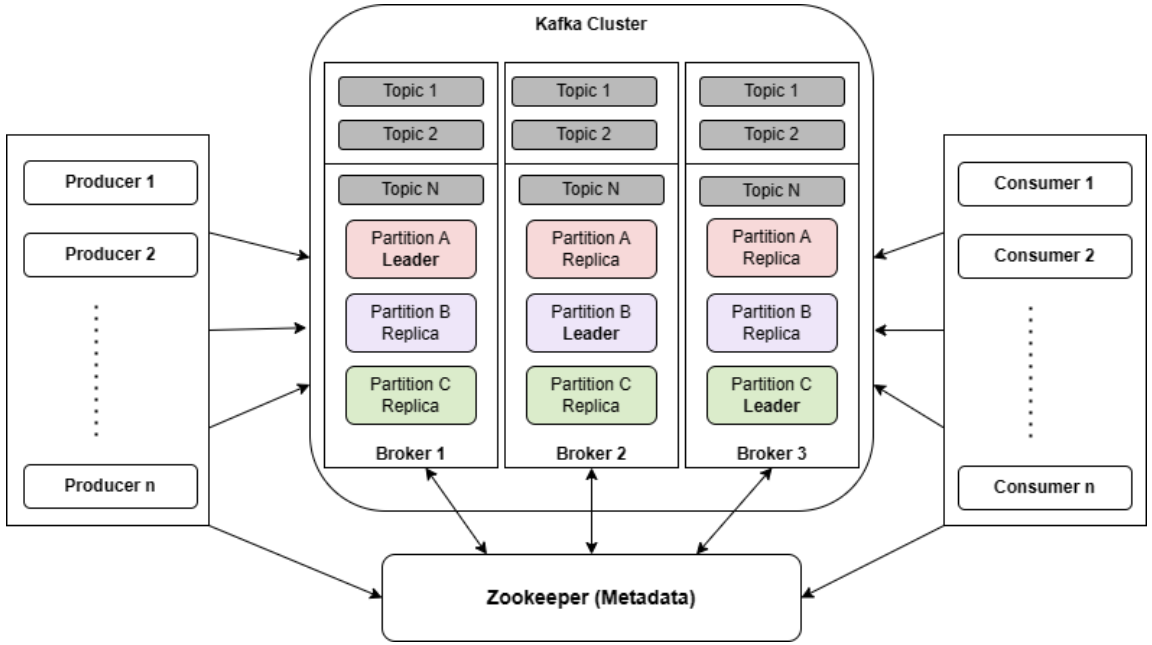
ZooKeeper and KRaft

- Cluster metadata storage
  - Kafka used ZooKeeper for coordination:
    - ZooKeeper stored cluster metadata
    - ZooKeeper hosted the controller election
    - Required running a separate ZooKeeper ensemble alongside Kafka
  - Modern Kafka uses KRaft (Kafka Raft):
    - Metadata stored in Kafka itself, as a replicated log
    - Controller election handled by a Raft consensus protocol
    - No ZooKeeper required
    - KRaft is the default for new clusters
  - Both are commonly found in production systems



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Producers and Consumers

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The Producer

- Kafka supplies a Producer API
  - Defines an internal process pipeline
  - When the application calls send(record), the producer
    - Serializes the key and value using the configured serializers
    - The partitioner decides which partition the record should land in
    - The record is appended to an in-memory buffer called the record accumulator, organized by partition
    - A separate sender thread continuously drains this buffer, batching records together by destination broker, and sends them over the network.
    - The application-facing send() call returns as soon as the record is accepted into the buffer
    - Returns a Future
    - Actual acknowledgement from the broker comes later, and you retrieve it through the future

```
[Application]
     | producer.send(record)
     ▼
[Serializer]          — turn key and value into bytes
     |
     ▼
[Partitioner]         — decide which partition
     |
     ▼
[Record Accumulator]  — buffer records, batch by partition
     |
     ▼
[Sender Thread]       — send batches to brokers
     |
     ▼
[Broker]              — writes, replicates, acknowledges
```

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The Producer

- Producer API parameters
  - Batching: grouping records going to the same partition into a single request
    - Key setting: batch.size (default 16 KB)
  - Linger: waiting briefly to let more records accumulate before sending
    - Key setting: linger.ms (default 0)
  - Compression: compressing the batch before sending
    - Key setting: compression.type (none, gzip, snappy, lz4, zstd)
    - Larger batches gives better compression ratios
    - Higher linger means larger batches which means higher throughput but higher latency
    - Compression means less network, less storage, more CPU

```
[Application]
     | producer.send(record)
     ▼
[Serializer]          — turn key and value into bytes
     |
     ▼
[Partitioner]         — decide which partition
     |
     ▼
[Record Accumulator]  — buffer records, batch by partition
     |
     ▼
[Sender Thread]       — send batches to brokers
     |
     ▼
[Broker]              — writes, replicates, acknowledges
```

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Send Modes

- The producer's send() method
  - Is asynchronous under the hood
  - It returns a CompletableFuture regardless of how you use it
  - The "mode" is really about what you do with that future
  - Fire-and-forget means you don't look at it
    - You accept that you won't notice failures
    - This is appropriate for streams where occasional loss is acceptable and the volume makes per-record handling impractical
    - For example, telemetry, metrics, application logs

Mode	How	When to Use
Fire-and-forget	`kafkaTemplate.send(topic, key, value)` and ignore the returned future	High-volume, loss-tolerant streams (telemetry, metrics, logs)
Asynchronous	`send(...)` and attach a `whenComplete()` callback to the returned `CompletableFuture`	Default for most production code — get delivery confirmation without blocking
Synchronous	`send(...).get()` to block the calling thread until the broker acknowledges	When the application genuinely cannot proceed until the event is durable

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Send Modes

- The producer's send() method
  - Asynchronous is the workhorse pattern for production code
    - You attach a callback so failures get logged, alerted on, or routed to a recovery mechanism
    - But the calling thread is never blocked
    - The producer pipeline keeps batching, the application keeps doing work, and you still find out when something goes wrong.
  - Synchronous is implemented by calling .get() on the future
    - Blocks the calling thread until the broker returns a response
    - Largely defeats Kafka's batching advantage
    - Use it only when the business semantics require it
    - For example, when a user-facing operation must not return success until the event is durably stored.

```java
// Fire-and-forget
kafkaTemplate.send("orders.placed", order.id(), event);

// Asynchronous (recommended default)
kafkaTemplate.send("orders.placed", order.id(), event)
    .whenComplete((result, ex) -> {
        if (ex != null) log.error("Send failed", ex);
    });

// Synchronous
SendResult<String, OrderEvent> result =
    kafkaTemplate.send("orders.placed", order.id(), event).get();
```

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ACKS

- The acks setting
  - Controls how many brokers must confirm a write before the producer considers it successful.
  - acks=0
    - The producer doesn't wait for any acknowledgment
    - It sends the bytes and moves on
    - If the broker rejects th
  - acks=1
    - The producer waits for the partition leader to write the record to its own log before acknowledging
    - This catches most failure modes like broker rejections, network errors, serialization issues
    - But if the leader crashes after writing locally but before any follower has replicated the record, the record is lost when a follower is promoted

Level	Who Must Acknowledge	Durability	Latency	When to Use
`acks=0`	Nobody — fire and forget at the wire	Lowest — silent loss possible	Lowest	High-volume, loss-tolerant streams (telemetry, metrics)
`acks=1`	The partition leader only	Medium — survives most failures, loses data if leader crashes before replication	Medium	Throughput-sensitive workloads where some loss is acceptable
`acks=all` (or `-1`)	Leader + all in-sync replicas	Highest — survives any single broker failure	Highest	Important business data; the production default

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ACKS

- The acks setting
  - acks=all
    - The producer waits for the leader and all in-sync replicas to persist the record
  - The latency cost grows with each level
    - acks=all requires waiting for replication, which adds milliseconds depending on the cluster
    - For most enterprise scenarios, that cost is well worth the guarantee
    - Modern Kafka makes acks=all with idempotence the default precisely because it's the optimal answer for almost all use cases

Level	Who Must Acknowledge	Durability	Latency	When to Use
`acks=0`	Nobody — fire and forget at the wire	Lowest — silent loss possible	Lowest	High-volume, loss-tolerant streams (telemetry, metrics)
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`acks=all` (or `-1`)	Leader + all in-sync replicas	Highest — survives any single broker failure	Highest	Important business data; the production default

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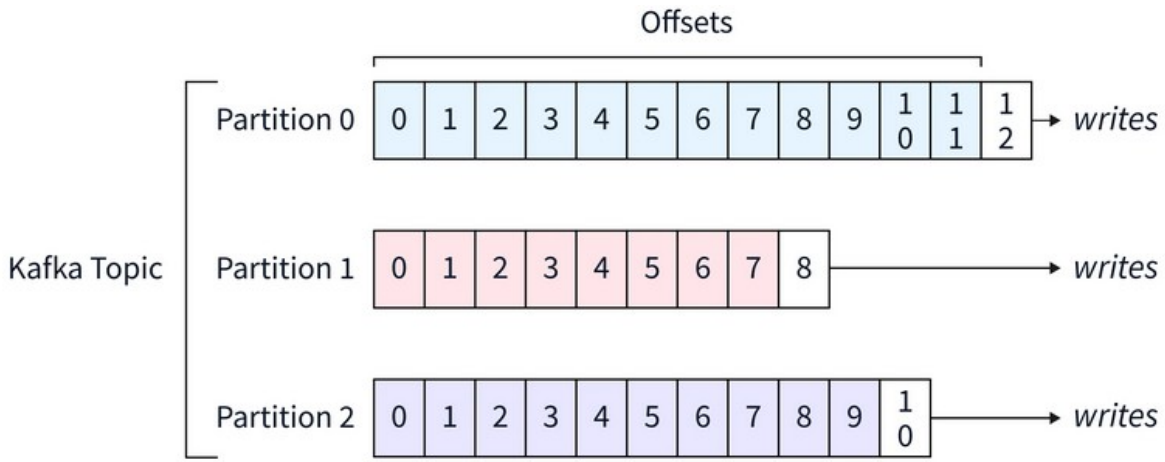
Partitioning Strategies

- Assigning records to partitions
  - If the application specifies a partition explicitly, then use it
  - Else if the record has a key
    - Hash the key, modulo partition count
    - Same key maps to the same partition
    - Preserves ordering for events with the same key
  - Else (no key)
    - Pick one partition, send records to it until the batch fills, then switch
    - Replaces older round-robin to improve batching efficiency
  - Custom partitioners are possible but rarely needed



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Idempotent Producers

- Network failures can cause producers to retry
  - Without checking, a retry might cause a duplicate:
    - Original request succeeded
    - Acknowledgment was lost
    - Producer retries
    - Broker receives the same event twice
  - Idempotent producers eliminate this:
    - Each producer is assigned a unique producer ID
    - Each message gets a per-partition sequence number
  - Broker rejects duplicate sequence numbers
    - Default in modern Kafka

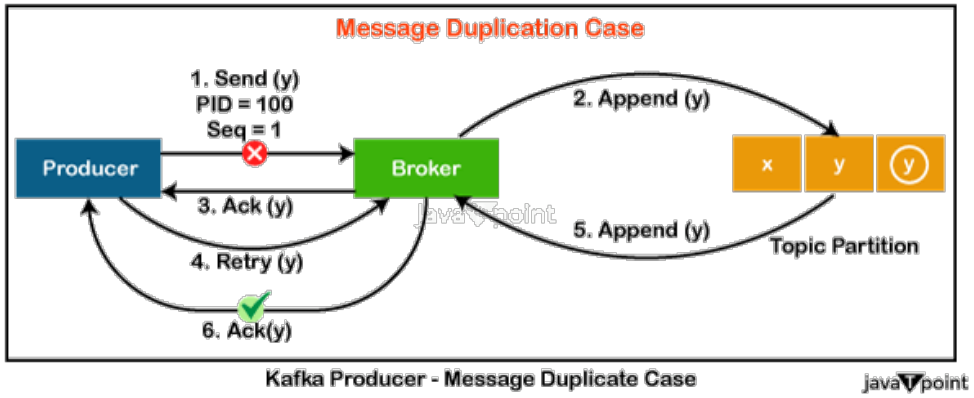

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Image Credit: https://www.tpointtech.com/apache-kafka-idempotent-producer

Consumer

- Built around a single polling loop
  - Every poll() call:
    - Sends heartbeats to the group coordinator
    - Fetches events from assigned partitions
    - Triggers rebalance handling if needed
    - Returns a batch of records to the application
  - The poll call is also the consumer's liveness signal
  - Failure to poll over a timeout period
    - Interpreted by the Broker that the consumer has terminated
    - Triggers a rebalance
    - The dead consumer’s partitions are assigned to another consumer in the same consumer group

```
while (running):
    records = consumer.poll(timeout)
    process(records)
    consumer.commitSync()   // or async
```

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Consumer Groups

- Consumers in the same group share the topic's partitions
  - Kafka assigns each partition to exactly one consumer in the group
  - Assignment is done the Kafka lead broker
- Assignment strategies
  - Range (default): contiguous partitions per consumer
  - Round-robin: alternating partitions across consumers
  - Sticky: minimize movement on rebalance

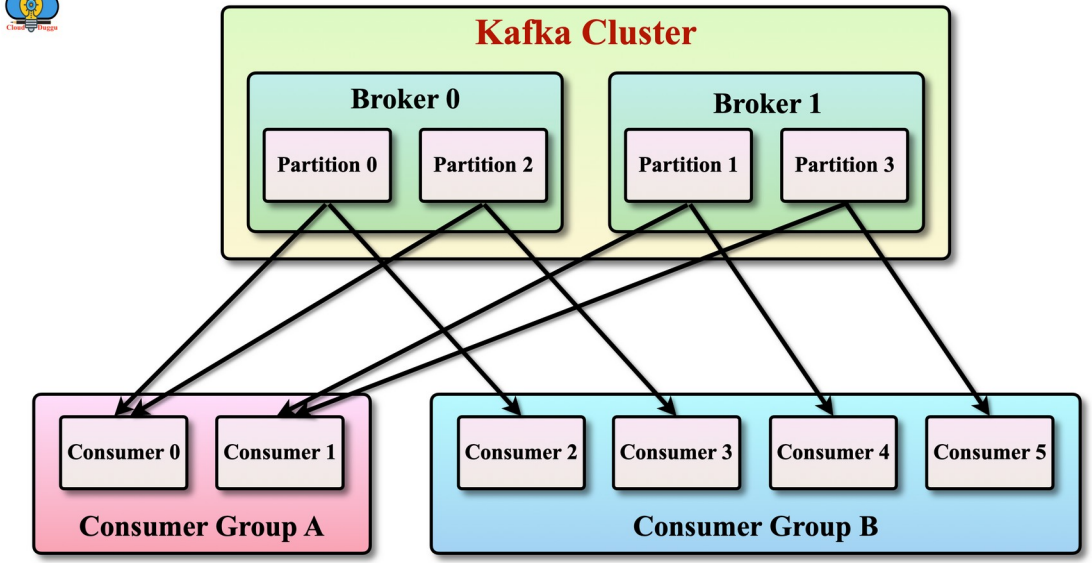

Image Credit: https://www.cloudduggu.com/kafka/consumers/r

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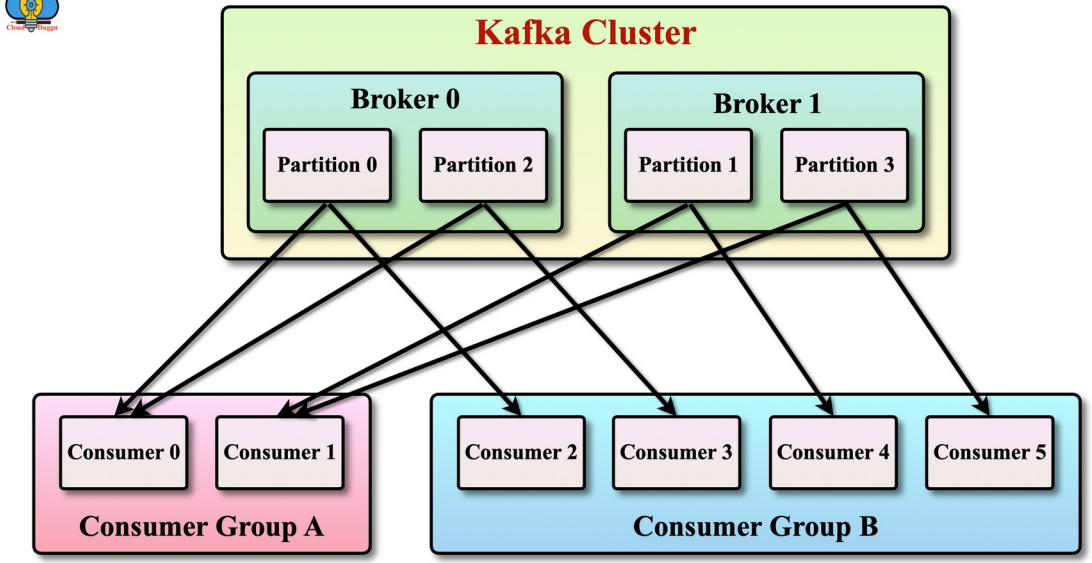
Consumer Groups Rebalance

- A rebalance redistributes partitions across consumer group members
  - Triggered by
    - A new consumer joining the group
    - An existing consumer leaving (gracefully or via failure)
    - A change in topic subscription
    - Partition count increasing for a subscribed topic
  - Classic rebalance ("eager" rebalance)
    - All consumers stop processing
    - New assignment is computed
    - All consumers resume with their new partitions
  - Cooperative rebalance
    - Only the partitions being reassigned are paused
    - Other processing continues



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Design Considerations

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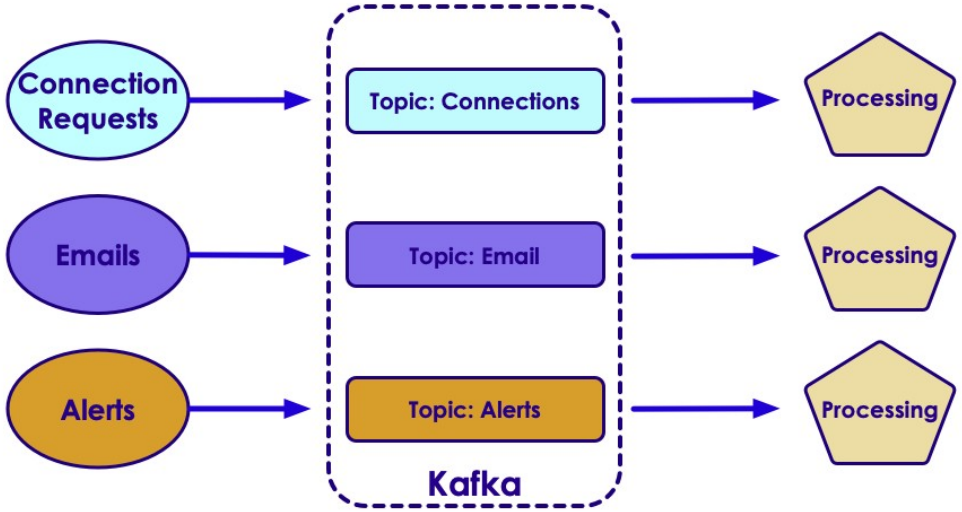
Topic Design

- Topic Ownership
  - In a multi-team organization, topics without clear ownership become liabilities
  - Ownership means
    - This team decides the event schema
    - This team is responsible for schema evolution
    - This team is on call for issues with this topic
    - Consumers know whom to contact for changes or clarifications
    - Document ownership in a topic registry or data catalog
  - Without ownership, you get "topic rot"
    - Stale message contracts nobody will fix



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Topic Design

- Costs of badly designed topics.
  - Renaming a topic requires consumer migration
  - Changing partition count disrupts key-based ordering
  - Changing cleanup policy between delete and compact means complicated migration
  - Changing replication factor mean cluster-level operation with performance impact
- Design topics as if they were APIs, because they are

Setting	What It Controls
`partitions`	Parallelism and ordering scope
`replication.factor`	Durability and fault tolerance
`retention.ms`	How long events are kept (time-based)
`retention.bytes`	How large the partition can grow before cleanup
`cleanup.policy`	`delete` (default) or `compact`
`min.insync.replicas`	Minimum replicas that must ack a write
`compression.type`	Broker-level compression policy

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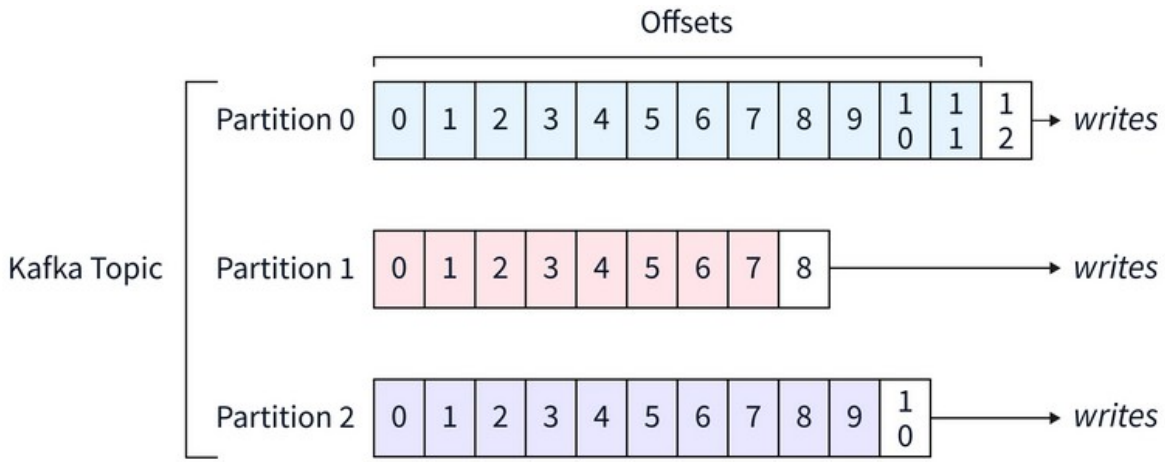
Partitioning Strategy

- What partitioning controls
  - Parallelism
    - More partitions = more parallel consumers possible
  - Ordering
    - Ordering is preserved within a partition, not across
  - Distribution
    - Partitions spread storage and load across brokers
  - These three concerns sometimes pull in different directions
  - Partition count and key selection are how you balance them



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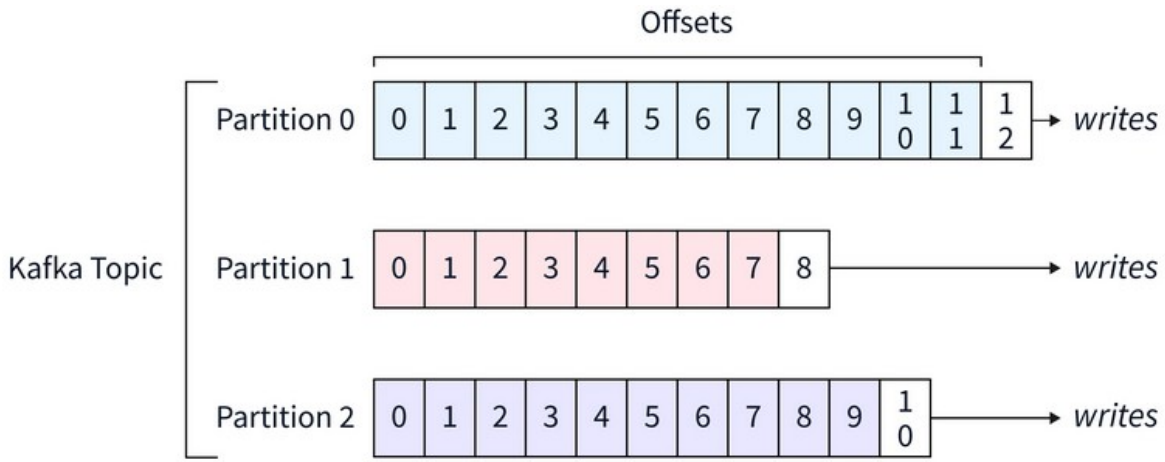
Partitioning Strategy

- Choosing a partition count
  - Partition count is a capacity and parallelism decision
  - Lower bounds
    - At least as many as the maximum expected consumer count per group
    - Enough to distribute load across brokers
  - Upper bounds
    - More partitions = more file handles, more memory, longer rebalances
    - More partitions = more metadata for the cluster to manage
  - Rule of thumb: start with enough partitions to cover expected peak parallelism, plus some headroom



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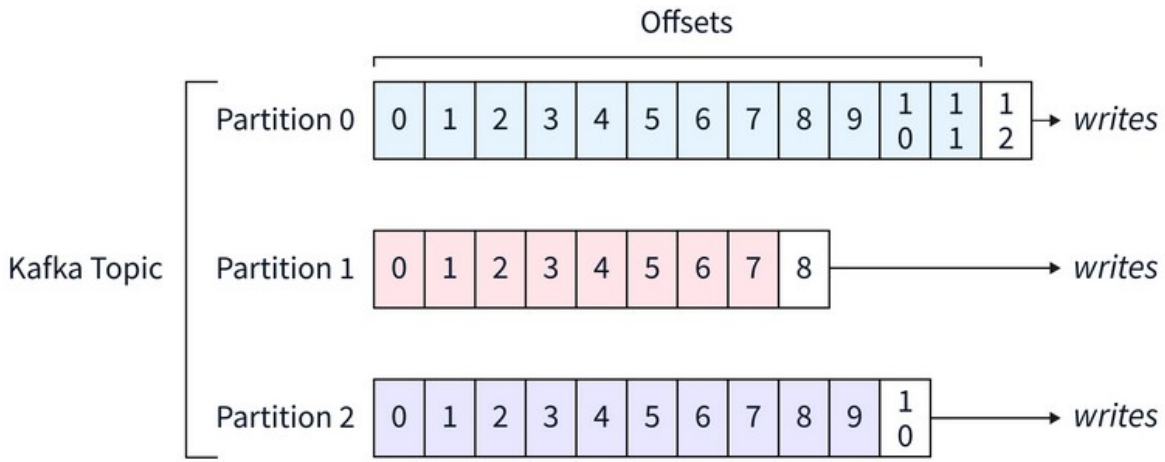
Partitioning Strategy

- Changing a partition count
  - You can add partitions to an existing topic
    - But the key-to-partition mapping changes
    - Events with key "customer-42" that used to go to partition 3 might now go to partition 7
    - Historical events remain in their original partitions
    - Consumers that depend on per-key ordering break silently
  - You cannot reduce partition count at all



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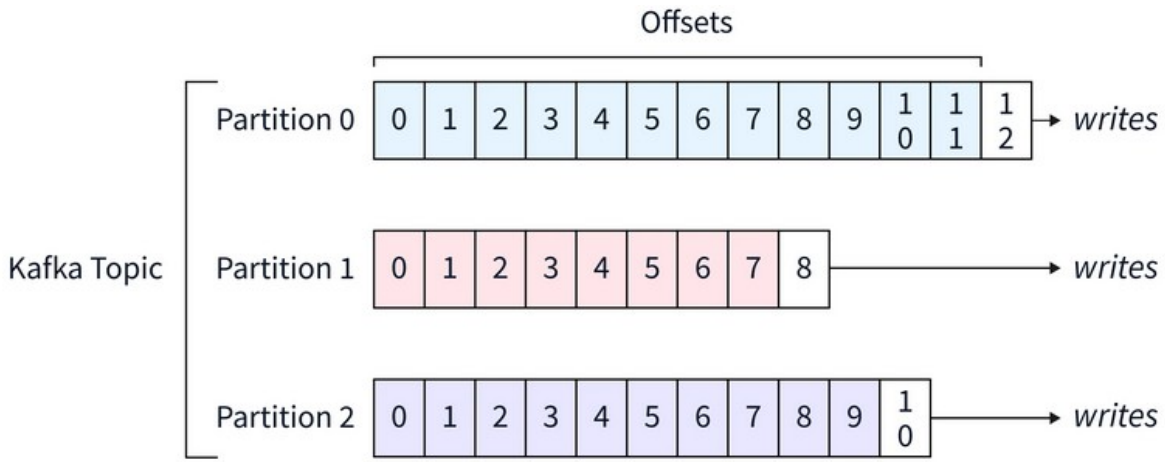
Partitioning Strategy

- Choosing a partition key
  - The key determines which partition an event goes to
- Events with the same key are ordered relative to each other
- Events with different keys have no ordering guarantee
- Good key = the "thing" whose lifecycle you want ordered:
  - For order events: the order ID
  - For user activity: the user ID
  - For account transactions: the account ID
- Bad keys are often too granular or too coarse



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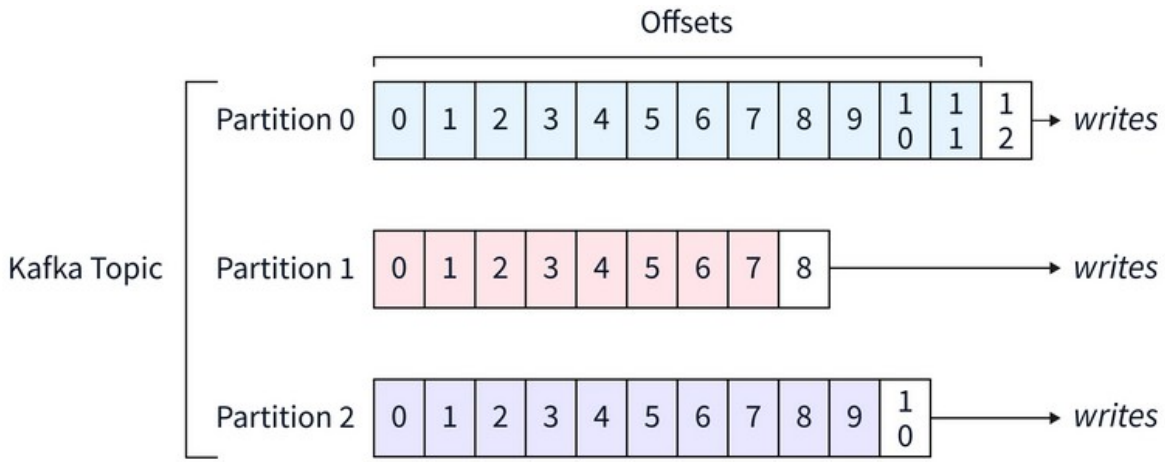
Partitioning Strategy

- When to skip the key
  - Not every event needs a key
  - No key means the producer uses a partitioning algorithm
- Good fit when
  - Events are independent of each other
  - No per-entity ordering is required
  - Maximum throughput matters more than ordering
  - Examples
    - Log streams
    - Metrics samples
    - Clickstream events



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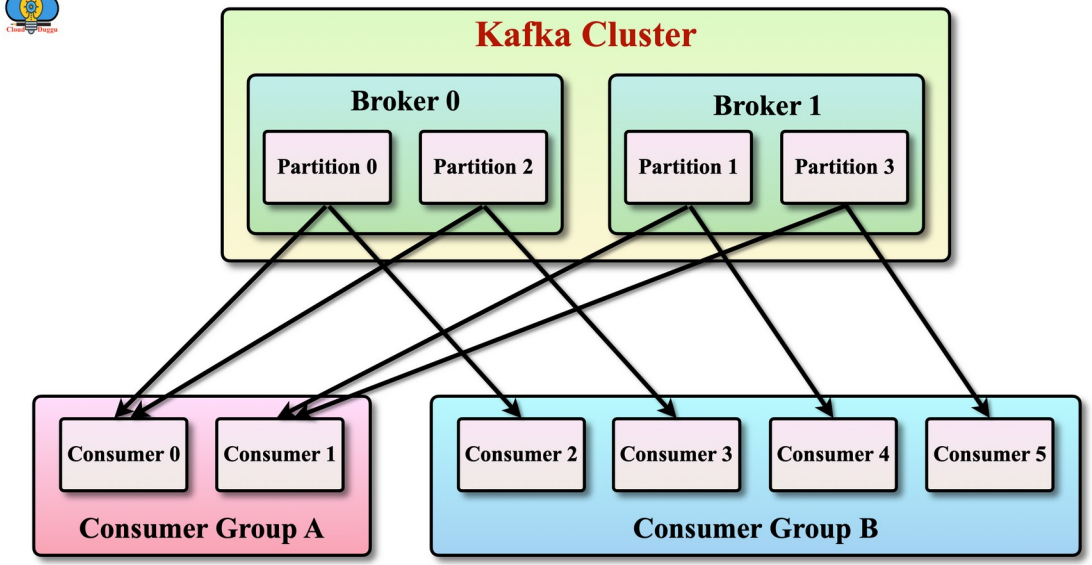
Consumer Groups

- The consumer group rule
  - One partition, one consumer (per group)
  - Within a consumer group, Kafka assigns each partition to exactly one consumer
    - Parallelism within a group is capped at the partition count
    - Adding a 13th consumer to a group reading a 12-partition topic means the 13th sits idle
  - Across groups, no such constraint
    - Each group gets its own copies of the messages



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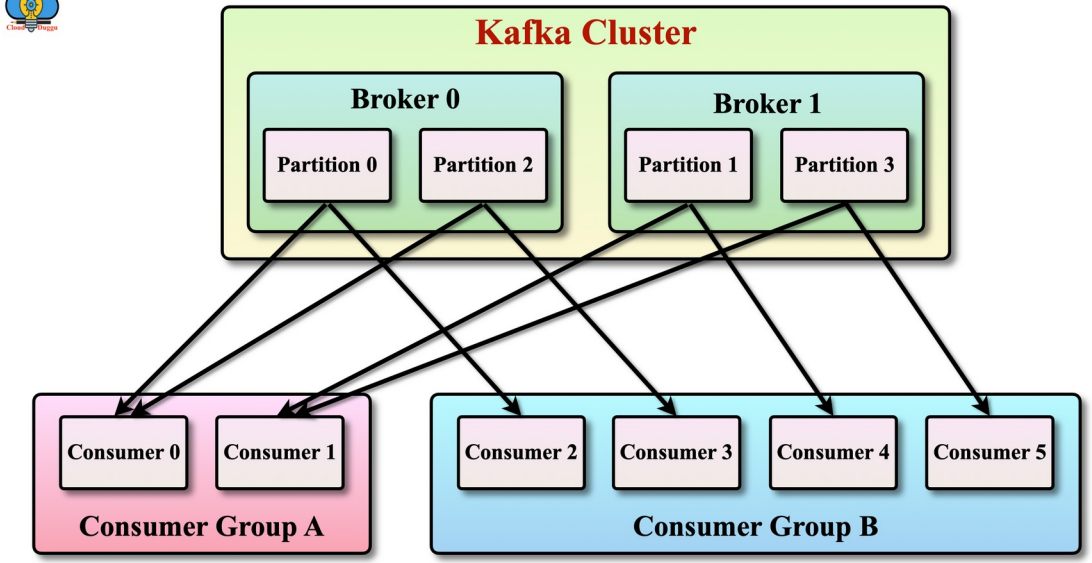
Consumer Groups

- Sizing a consumer group
  - Main question: what's the processing rate you need vs. the production rate?
  - Factors
    - Peak event arrival rate
    - Average processing time per event
    - Acceptable consumer lag
  - Start point
    - Required_consumers = (peak_rate × avg_processing_time) / target_utilization
  - Then cap at partition count



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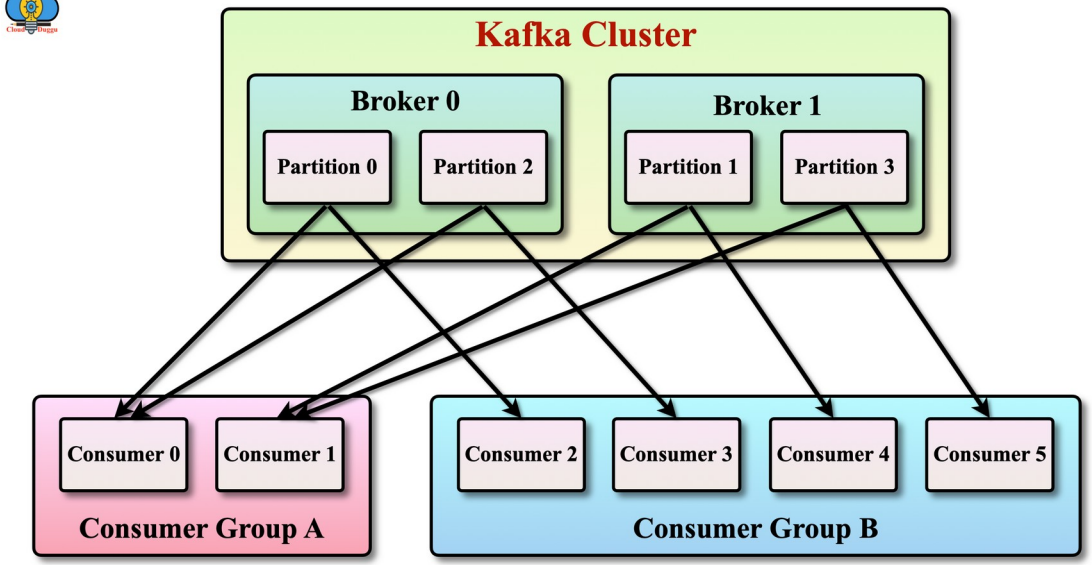
Consumer Groups

- Number of consumer groups
  - A single topic can support many consumer groups simultaneously
  - Each group has its own
    - Offset positions
    - Membership and coordination
    - Processing logic
  - Groups are a fan-out
    - Each group represents a domain of concern
  - Orders topic might be read by independent interests
    - Inventory services
    - Accounting services
    - Warehouse management services
  - Each interest would have its own consumer group
  - New consumer groups are additive and have zero impact on existing ones



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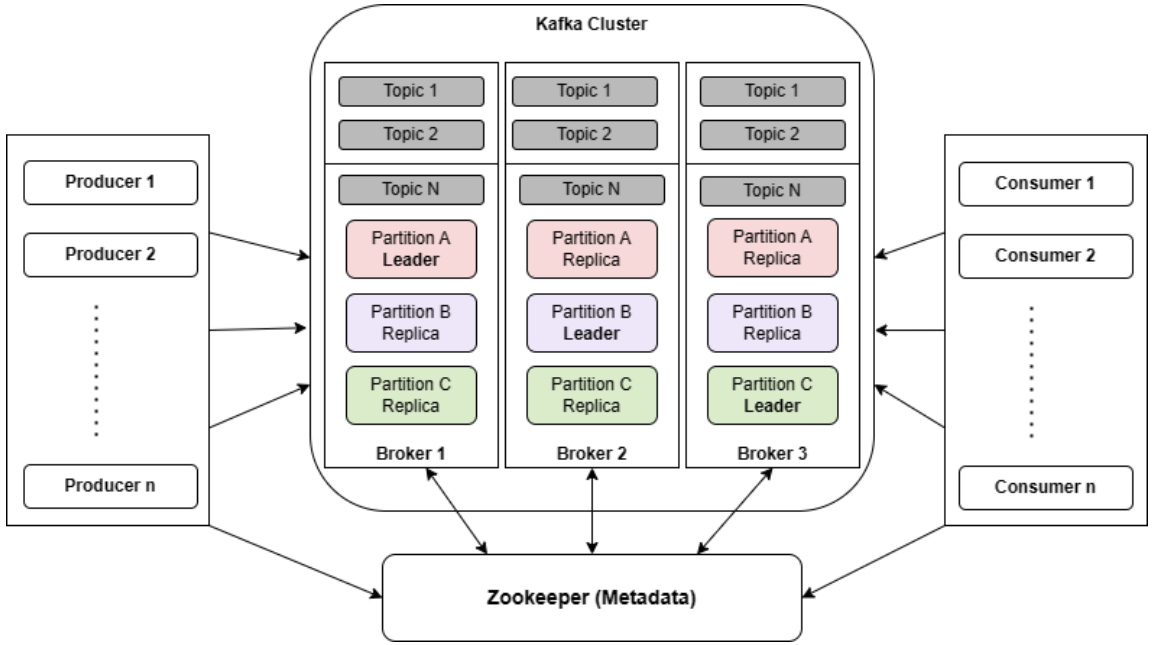
Topic Design Checklist

- When designing a topic
  - Topic name follows the naming convention
  - Topic ownership is documented
  - Granularity (per event, per aggregate, per context) is deliberate
  - Partition count accounts for peak parallelism and growth
  - Partition key is chosen with ordering requirements in mind
  - Known hot-key risks have been considered
  - Retention matches consumer replay needs
  - Replication factor and min.insync.replicas are set for durability requirements
  - Consumer groups and their purposes are enumerated



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Questions?